



Digital IC Design

数字集成电路设计

Institution of VLSI Design, College of Integrated Circuits

朱晓雷 (Xiaolei Zhu, PhD), 陈泰海 (Taihai Chen, PhD)

Xl_zhu@zju.edu.cn

2024

*Some Contents Adapted from Copyright Prentice Hall/Pearson, Rabaey's Digital Integrated Circuits, Mary Jane Irwin's VLSI Digital Circuits and University of Bolton, Weste's CMOS VLSI Design



Self Introduction



❑ XIAOLEI ZHU

- PhD., Associate Prof./ PhD Adviser.
- Institution of VLSI Design, College of Integrated Circuits

❑ Education:

- the M.S. degree in electrical engineering from Zhejiang University, China, in 2003,
- the Ph.D. degree in electrical engineering from Keio University, Japan, in 2012.

❑ Working Experience:

- From 2003 to 2006, he was with the Institute of Micro-Electronics, Tsinghua University, China, as an Assistant Researcher.
- From 2007 to 2012, he was with Fujitsu Laboratories Ltd., Kawasaki, Japan, where he developed high performance mixed-signal integrated circuits and system.
- Since 2013, he has been with the Institute of VLSI design, Zhejiang University, as an Associate Professor.

- ❑ **Current research interests:** include LSI implementation for in memory computing, the neuromorphic computing and Chiplet integration as well.



Chapter 1

Preface & Manufacture & Device



Course goals

- Introduction to **CMOS digital** integrated circuits
 - Objects of VLSI design(Technology, MOS Circuit, Sequential Logic Circuit, Module and Unit, RTL Design, Architecture)
 - Methods of VLSI design(Coding, Test and Verification, Synthesis)
- Course goals
 - Basic ability to design and implement CMOS circuits and optimize them with respect to different constraints: **size, cost, speed, power dissipation**, and **reliability**
 - For beginners of VLSI design
 - To the objects of VLSI Design → How to evaluation
 - To the methods of VLSI Design → How to manager



Course Contents

- Basic circuit theory
 - resistance, capacitance, inductance
 - MOS gate characteristics
- Logic design
 - logical minimization, FSMs, component design
- Hardware description language *
- VHDL or verilog
- Use of modern EDA tools *
- simulation, synthesis, validation (Synopsys)
- schematic capture tools



Materials

- **Reference text book**

- Jan M. Rabae, 数字集成电路-电路、系统与设计(第2版), /Digital Integrated Circuits: A Design Perspective;
- Stephen Brown and Zvonko Vranesic, Fundamentals of Digital Logic with Verilog Design (Third Edition).

- **Other Reference materials**

- Neil H.E. Weste et.al. CMOS超大规模集成电路设计 (第4版) , / CMOS VLSI Design A Circuit and System Perspective 4th Edition .
- Wayne Wolf, Modern VLSI Design: IP-Based Design (4th Edition) , Prentice Hall;
- Text PPT。

- **Journals and proceedings**

- **IEEE Journal of Solid State Circuits;**
- **Proceedings of The International Solid-State and Circuits Conference (ISSCC);**
- **Proceedings of VLSI Symposium;**
- **European Solid-State Circuits Conference (ESSCIRC)**
- **Asian Solid-State Circuits Conference (ASSCC)**



Course Agenda

#	Date and Time	Topic	lecturer
1	12:30~15:30 PM, Dec. 24 (Tue)	Preface, Manufacturing process & Devices	Xiaolei Zhu
2	12:30~15:30 PM, Dec. 26 (Thur)	Combinational logic gates	Xiaolei Zhu
3	12:30~15:30 PM, Dec. 27 (Fri)	Verilog and Combination logic example	Taihai Chen
4	12:30~15:30 PM, Dec. 30 (Mon)	Sequential logic	Xiaolei Zhu
5	12:30~15:30 PM, Dec. 31 (Tue)	Verilog and Sequential logic example	Taihai Chen
6	12:30~15:30 PM, Jan. 1 (Wed)	Lab1	
7	12:30~15:30 PM, Jan. 2 (Thur)	Timing	Xiaolei Zhu
8	12:30~15:30 PM, Jan. 6 (Mon)	Arithmetic,Memory,Testing	Xiaolei Zhu
9	12:30~15:30 PM, Jan. 7 (Tue)	Dedicated Microprocessor	Taihai Chen
10	12:30~15:30 PM, Jan. 8 (Wed)	writing microprocessor with Verilog	Taihai Chen
11	12:30~15:30 PM, Jan. 9 (Thur)	Lab2	
12	11:00~13:00 PM, Jan. 10 (Fri)	Examination	Zhu & Chen



Course Administration

- Course Evaluation
 - Final Exam: 50%; Coursework & Lab: 50%;
 - Will adjust the score by the daily performance:
Contribution, Communication, Feedback
- Contact Info.
 - Xiaolei Zhu 朱晓雷
xl_zhu@zju.edu.cn; 13456999141
 - Taihai Chen 陈泰海
taihai.chen@zju.edu.cn



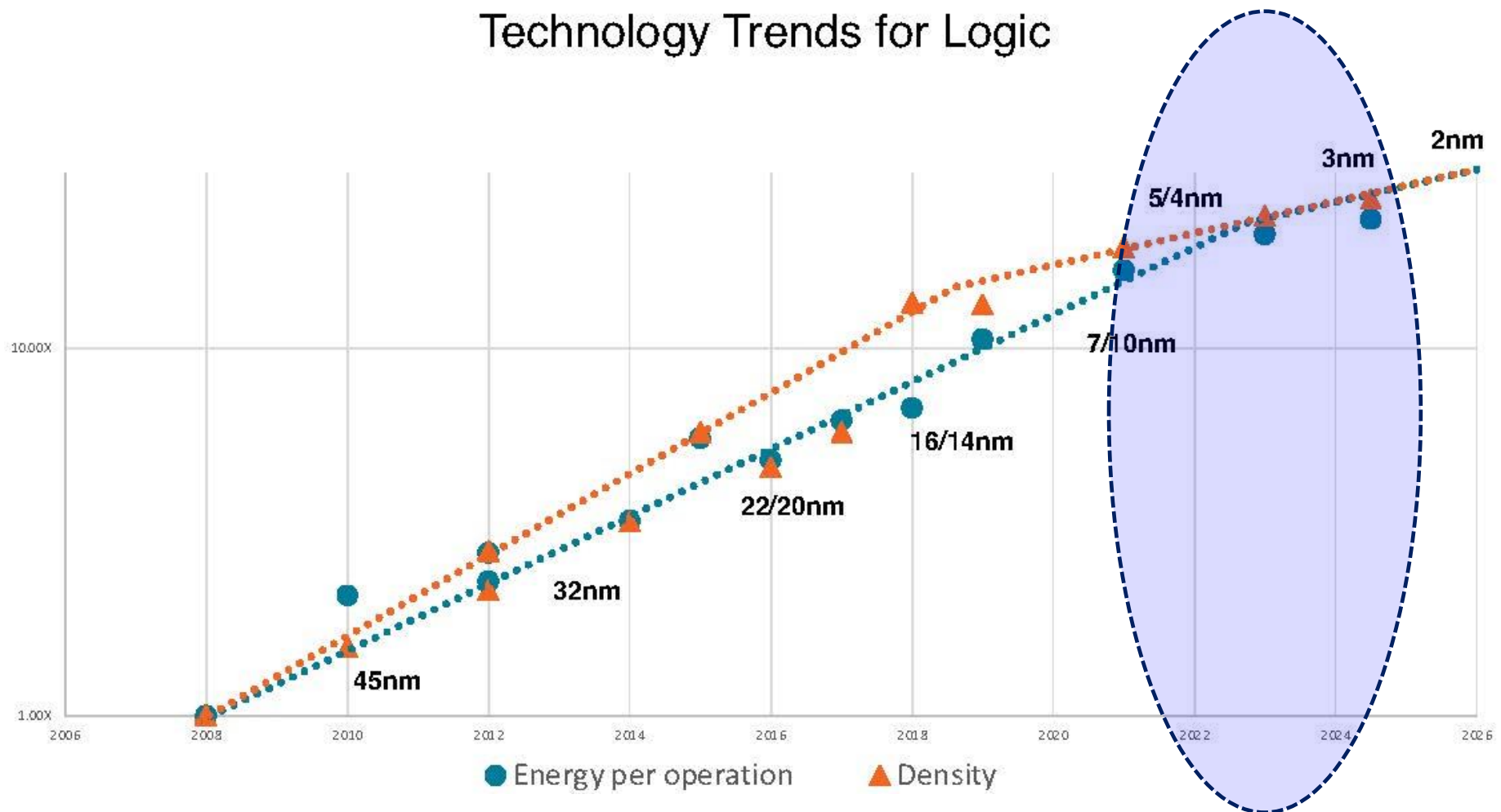


Brief Introduction of VLSI design

Where We Stand Now?

Lisa Su, AMD, [ISSCC 2023](#)

Technology Trends for Logic

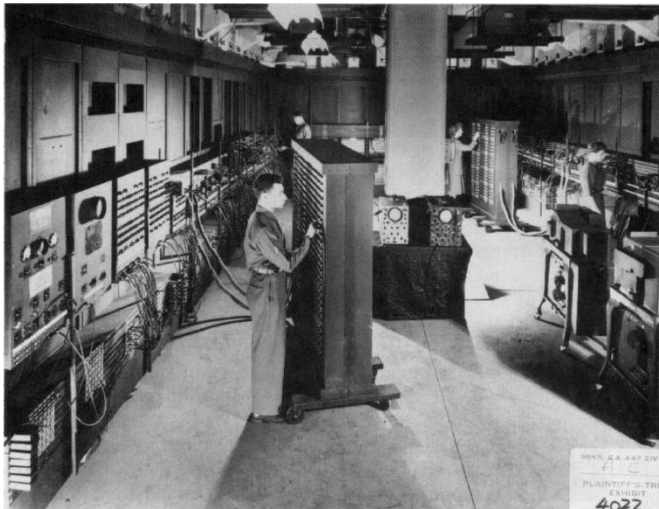


The Invention of Computer and Transistor

In 1939, Atanasoff in Iowa State University invented the world 1st computer "Atanasoff-Berry Computer";



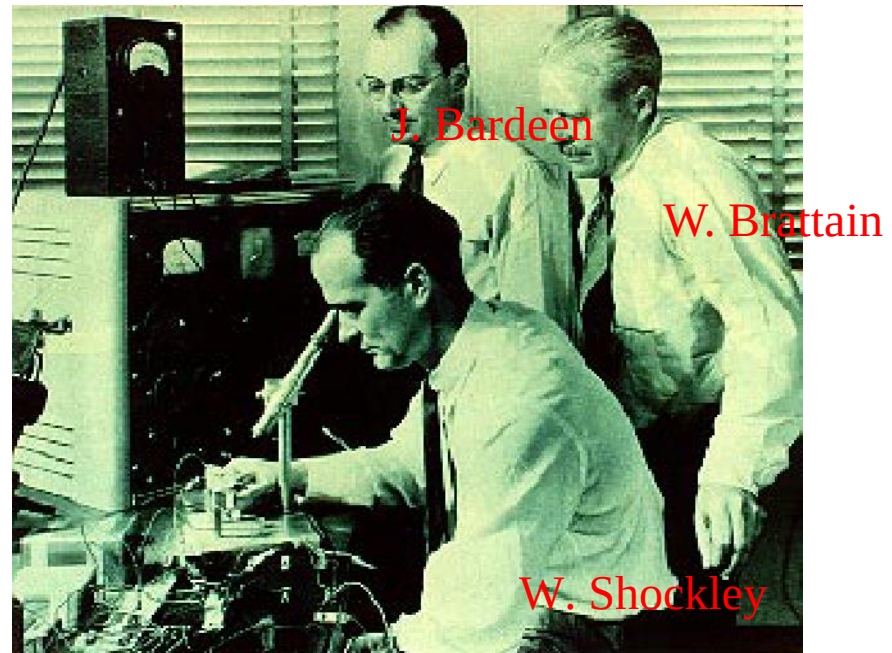
1946/02/14, in the University of Pennsylvania's Moore School of Electrical Engineering invented



Giant and Weak ENIAC(Electronic Numerical Integrator and Calculator)

The transistor was invented at 1947/12/23, in Bell Lab

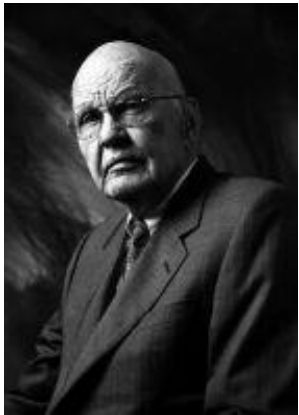
Won the Physics Nobel Prize in 1956



The Invention of IC

Clair Kilby invented the 1st IC in 1958 at TI Corporation, 12 components, Ge Semiconductor

Co-inventor, Robert Noyce, Planar Technology

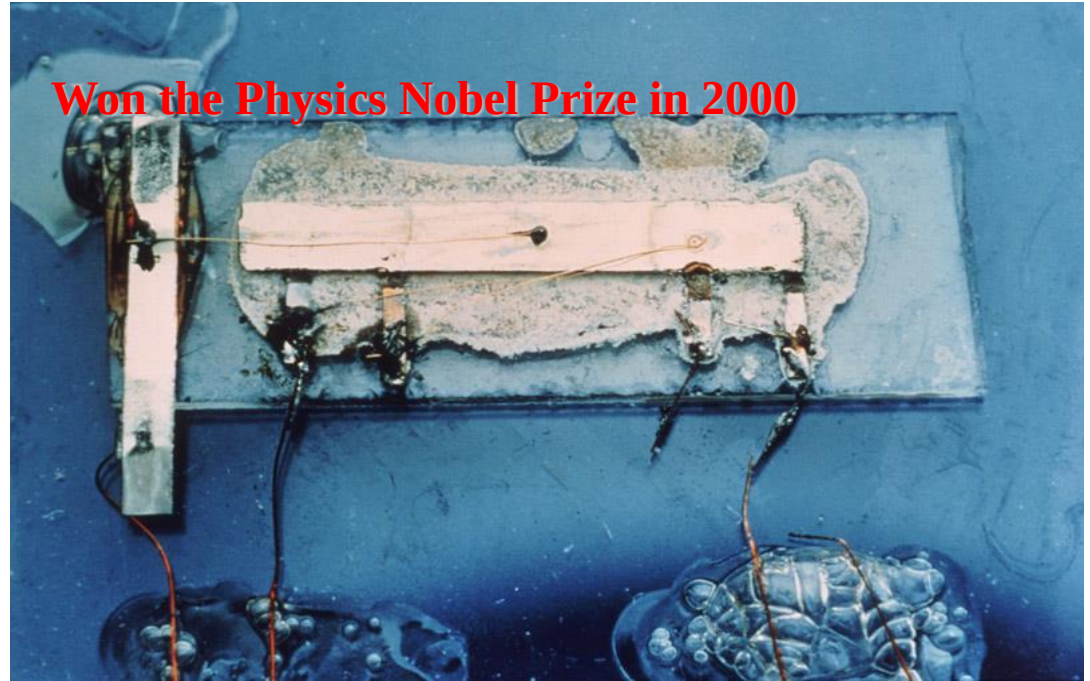


Kilby



Noyce

Won the Physics Nobel Prize in 2000



- The traitorous eight abandoned Shockley founding Fairchild Semiconductor.



Analog and Digital Chip in 1970's

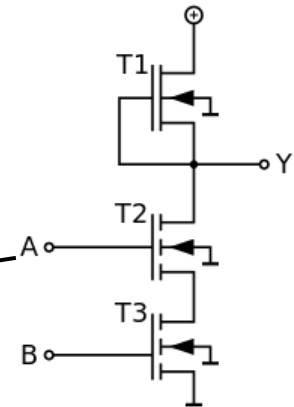
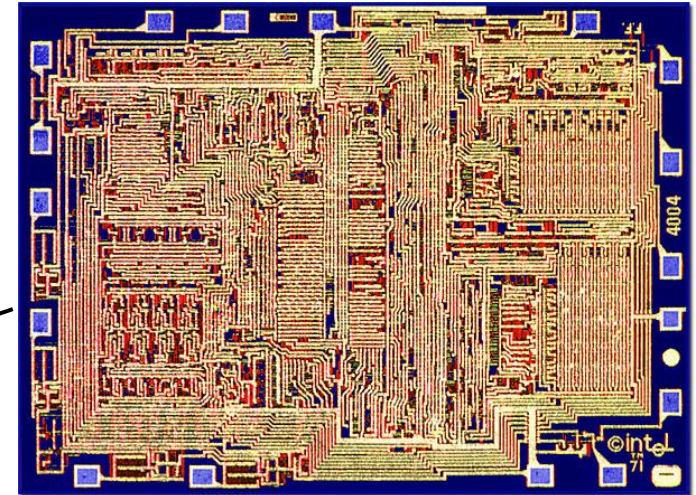
1960's: Analog chip-OpAmp

- Fairchild, μ A741, made of **BJT**

Intel CPU

- 4004 (1971) (10- μ m enhancement pMOS)
- 8085 (1976), 8-bit (address width: 16-bit),
3- μ m depletion pMOS, single $V_{dd} = 5V$
- 8086 (1978), 16-bit (a.w.: 20-bit), 3- μ m

Tsinghua Uni. @China: C μ 8085A (1981-1983) 有29,000个晶体管的16-bit CPU
C μ 8086 (1984-1987) 芯片设计与制作 (1984年获电子工业部科技进步一等奖)



A **depletion-load**
nMOS NAND gate



A Brief Review of IC/IT Development in Five Decades

1970-1979: MOS logical circuits, CPU

1980-1989: CMOS/memory, PC

1990-1999: CMOS scaling towards nanometer regime, networking/web-browser

2000-2009: Aggressive CMOS scaling in nm region (from 130nm to 32nm, new node every 2-years: 130, 90, 65, 45, 32) and Wi-Fi, smartphone, searching engine

2010-2019: Scaling slow down and EUV to rescue, Big data/Cloud computing, AI/DL

2020-: Post Moore era, AI/DL and ADAS (Advanced Driver Assistance System) , LLM



Driving force of IC design

The performance of the IC:

- Data processing and computing ability;
- Power.

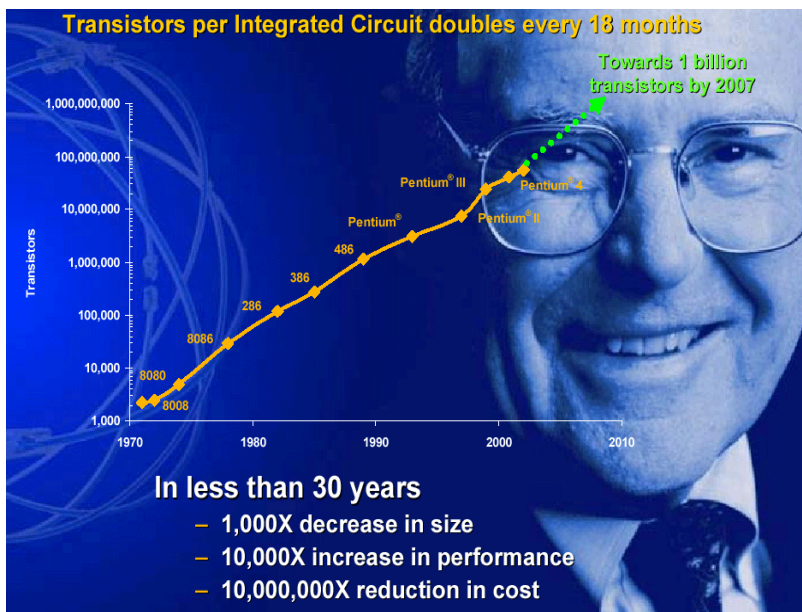
Price of the IC:

- Die cost;
- Package cost;
- Fabrication cost.

Timing for the market:

- Design cycle;
 - Fabrication cycle.
- 

Moore's Law to the Post Moore era



- Gordon Moore: co-founder of Intel.
- In 1965, predicted that number of transistors per chip would grow exponentially (double every 18 months [Later revise to 24 months]).

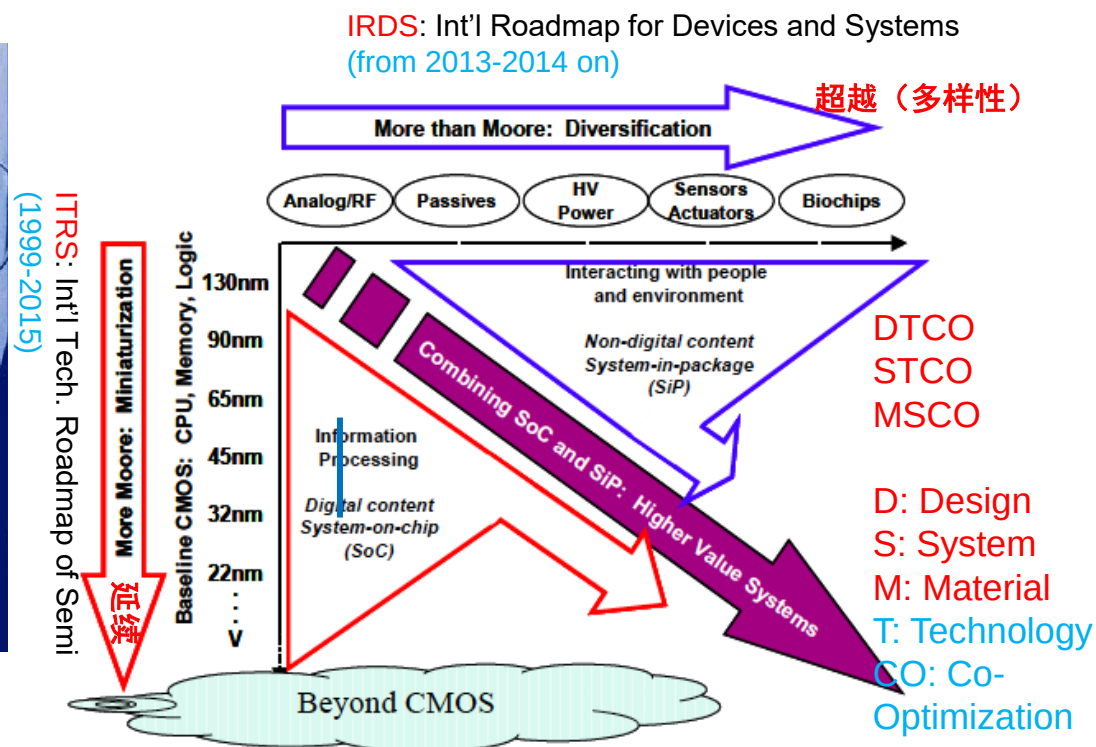


Figure 5 Moore's Law and More

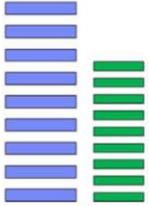
Two trends in the post Moore era:

- 延续摩尔 (More Moore)
- 超越摩尔 (More than Moore)

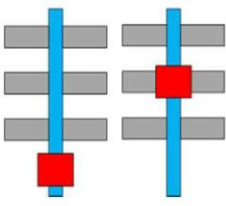


The Challenge for the Post Moore era

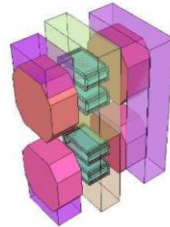
2D器件微缩



设计制造协同优化 (DTCO)



3D器件 (GAA/CFET)

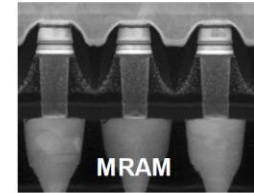


More transistors

堆叠SRAM

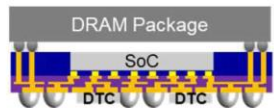


新型存储器

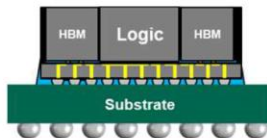


More memories

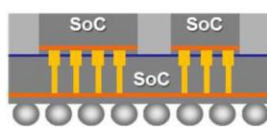
InFO



CoWoS®

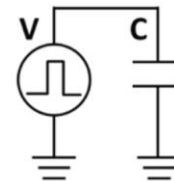


SoIC



Higher integration (Chiplets)

CV²

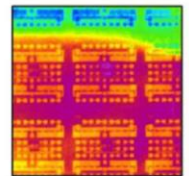


Domain-specific technology

Software
System
Architecture
Circuit
Device
Technology

Domain-specific architecture

Thermal Management

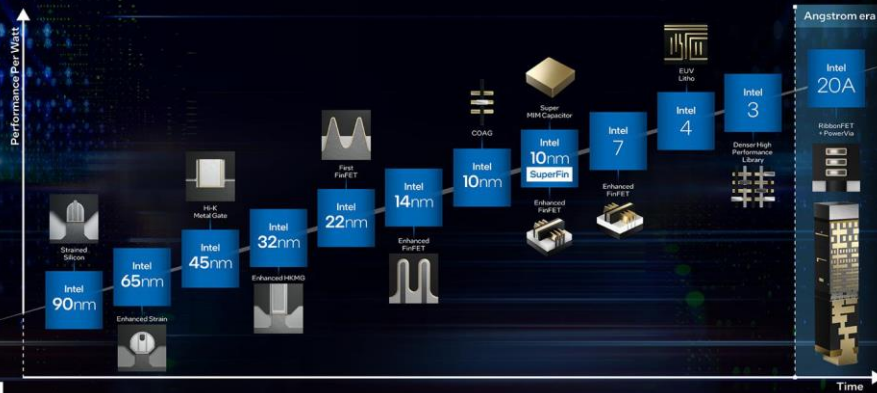


More advanced EDA tools



Development of the Device

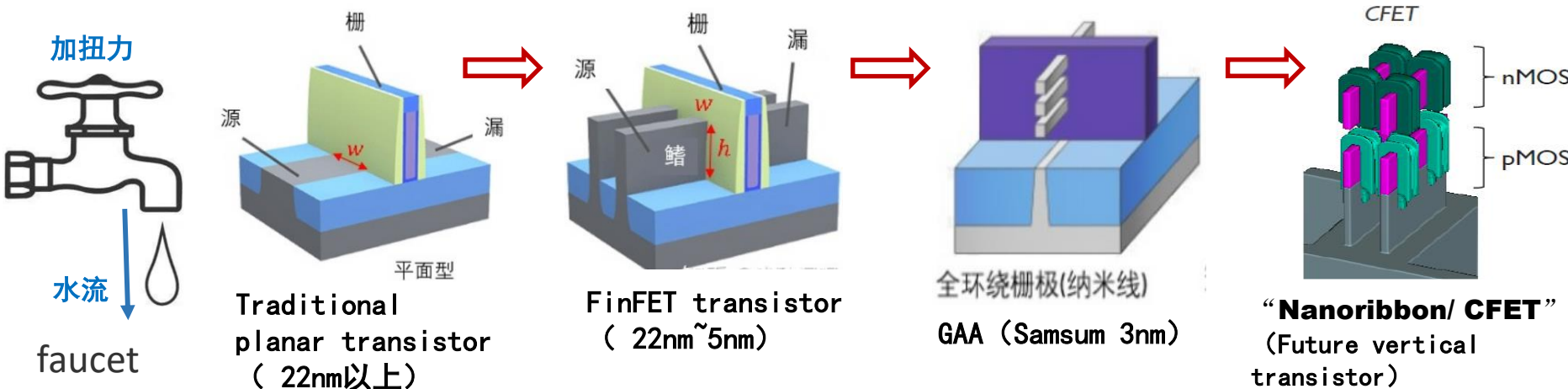
Intel Process Technology Innovations



Introducing Intel's new node naming



- 10nm SuperFin
 - 已大量生產
- Intel 7 (Formerly Enhanced SuperFin)
 - 每瓦效能約提升 10% ~ 15%
 - 2倍密度提升
 - 全面使用極紫外光 (EUV) 微影技術
 - 電晶體最佳化
 - 已量產
- Intel 4 (Formerly Intel 7nm)
 - 每瓦效能約提升 20%
 - 2倍密度提升
 - 全面使用極紫外光 (EUV) 微影技術
 - 更密的高速性能設計元件
 - 增加固有驅動電流
 - 降低接點阻值
 - 提升 EUV 使用比例
 - 2023 下半年開始生產
- Intel 3
 - 每瓦效能約提升 18%
 - 耗電及面積改善
- Intel 20A
 - 2024 上半年突破式創新
 - RibbonFET – 全新電晶體架構
 - PowerVia – 連線技術的創新





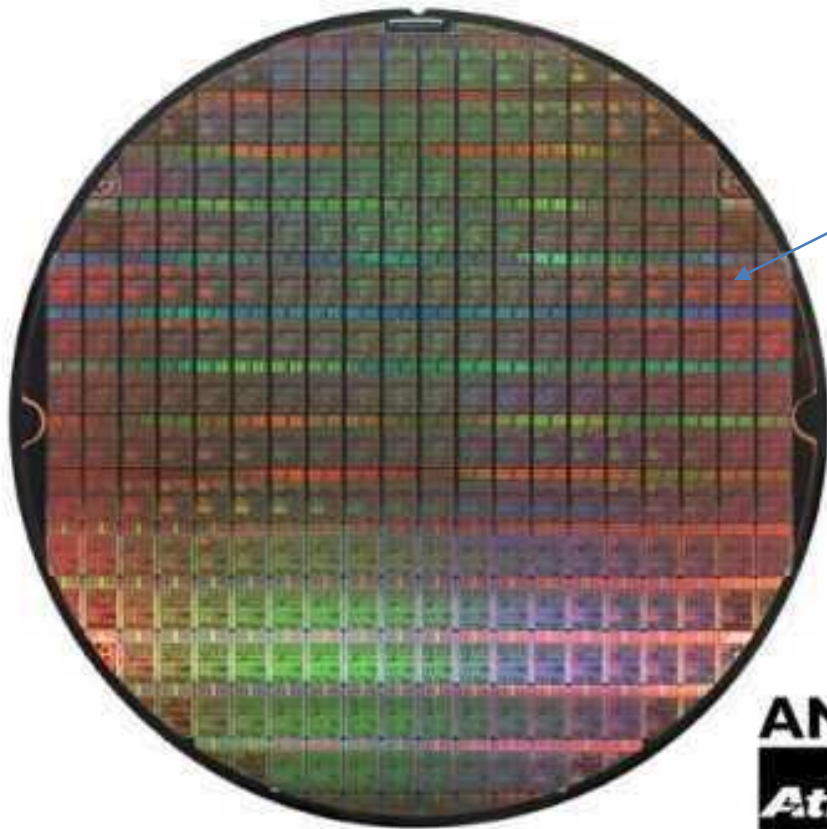
Cost of ICs

- NRE (non-recurrent engineering) cost or fixed cost
 - design time and effort, mask generation
 - one-time cost factor
- Recurrent cost or variable cost
 - silicon processing, packaging, test
 - proportional to volume
 - proportional to chip area

$$\text{Cost per IC} = \text{variable cost per IC} \\ + \text{fixed cost/volume}$$



Wafer and Die



Single die

Wafer



Going up to 12" (30cm)

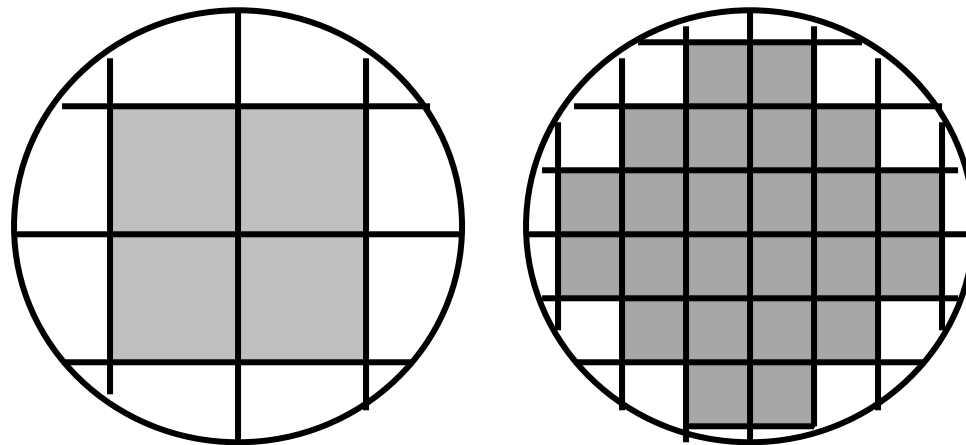


Yield

- $Y = \text{Yield}$
$$Y = \frac{\text{No. of good chips per wafer}}{\text{Total number of chips per wafer}} \times 100\%$$

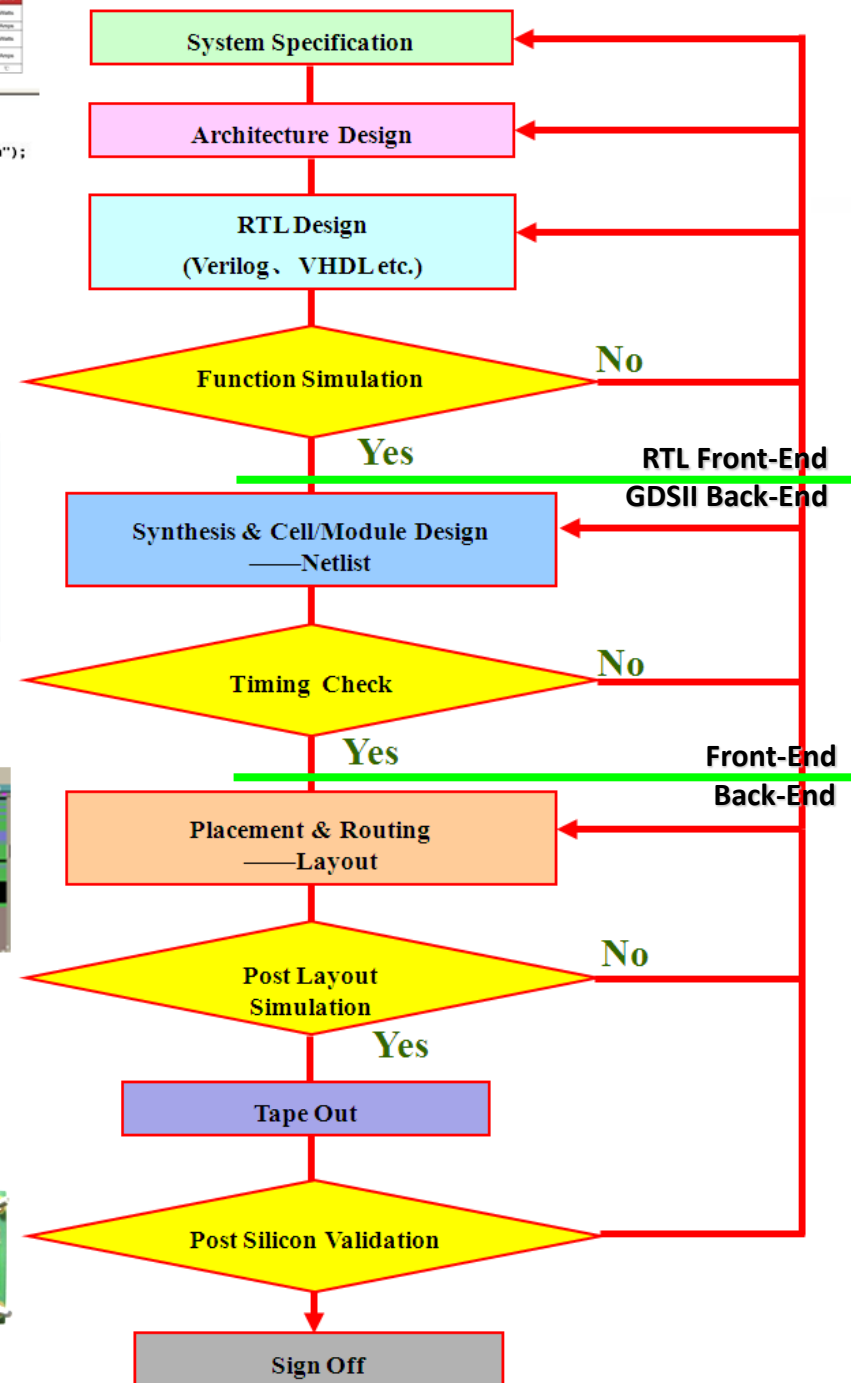
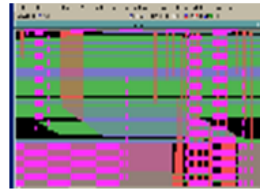
$$\text{Die cost} = \frac{\text{Wafer cost}}{\text{Dies per wafer} \times \text{Die yield}}$$

$$\text{Dies per wafer} = \frac{\pi \times (\text{wafer diameter}/2)^2}{\text{die area}} - \frac{\pi \times \text{wafer diameter}}{\sqrt{2} \times \text{die area}}$$





- | NAME | FORMAT | VALUE | UNIT |
|---|--------------|--------------|-------|
| Peak Power Power Dissipation at 1000Hz waveform | Power | Minimum 1000 | Watts |
| Power (Rated) [Peak] | Power | Minimum 1000 | Watts |
| Peak Power Current at 1000Hz waveform (Peak) [Fig.3] | Power | Min 500 | Amps |
| Steady State Power Dissipation at T_{amb} [1000Hz] (Steady State) [Fig.3] | Power | 0.5 | Watts |
| IR (Rated) (Max) | Power | 400 | Watts |
| Peak Forward Bias Current (Steady State) Single Half-Sine Wave Superequivalent on Rated Load [JEDEC] (Half-Peak) (Peak 3-Fig.3) | $I_{F, rms}$ | 400 | mAmps |
| Operating junction and Storage Temperature Range | $T_{j, max}$ | 200 to 1000 | °C |
-
- ```
#include<stdio.h>
main()
{
 printf("Hello World ! \n");
}
```





# Top-down vs. bottom-up design

- Top-down design adds functional detail.
  - Create abstractions of lower levels from upper levels.
- Bottom-up design generates cost from abstractions of lower levels.
- Good design needs both top-down and bottom-up efforts.





# THE MANUFACTURING PROCESS



# Fabrication services

Foundry: Fabrication line for hire.

- Pure-play foundries: TSMC, UMC, Global Foundries, SMIC, ...
- IDM (Integrated device manufacturers): Intel, Samsung, TI, NXP, Toshiba, Infineon, ST, ...

Fabless: Integrated device designers (no manufacturing, not foundries)

- Qualcomm, Broadcom, MediaTek, AMD, Marvell, Nvidia, ...



# Terminology

- Manufacturing node: technology at a particular channel length.
- Deep submicron technology: 250 nm and below.
- Nanometer technology: 100 nm and below.



# Fabrication processes

IC built on silicon substrate:

- some structures diffused into substrate;
- other structures built on top of substrate.

Substrate regions are doped with n-type and p-type impurities. (n+ = heavily doped)

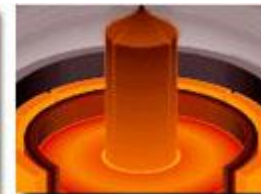
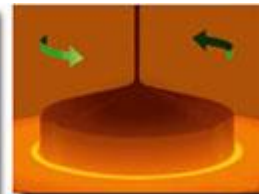
Wires made of polycrystalline silicon (poly), multiple layers of aluminum (metal).

Silicon dioxide ( $\text{SiO}_2$ ) is insulator.

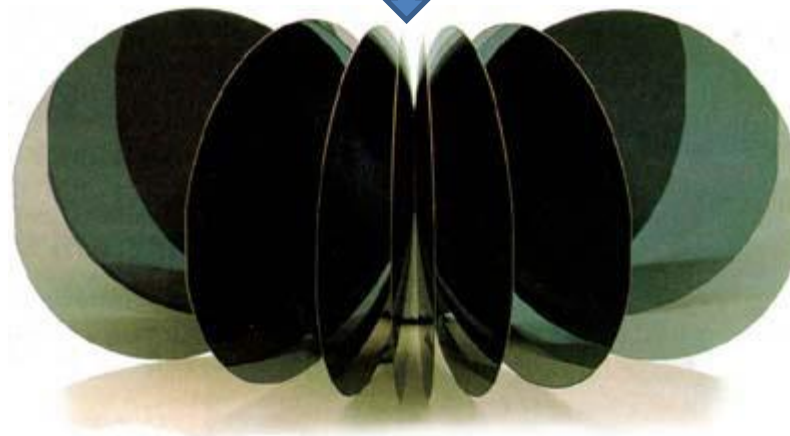




# Silicon Wafer



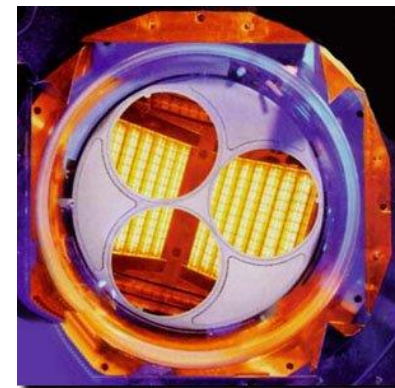
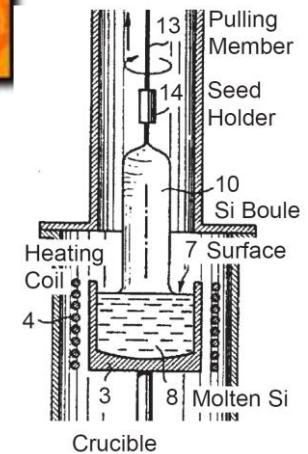
1ppb (part per billion)



4-12 inch

1mm

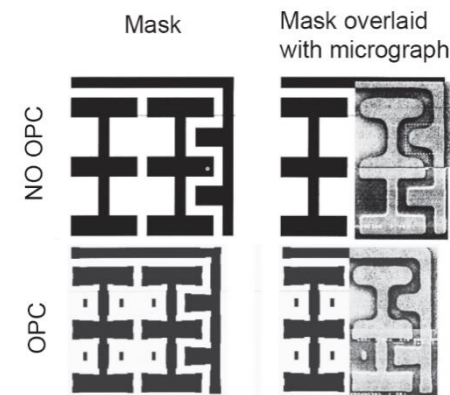
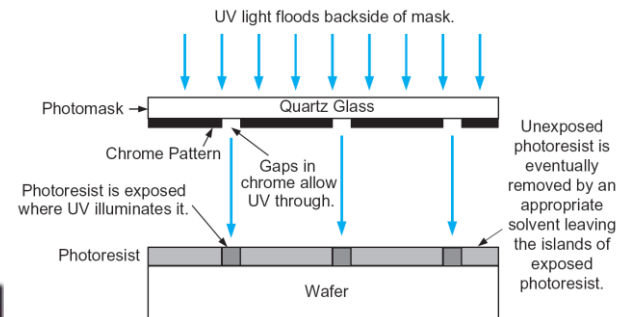
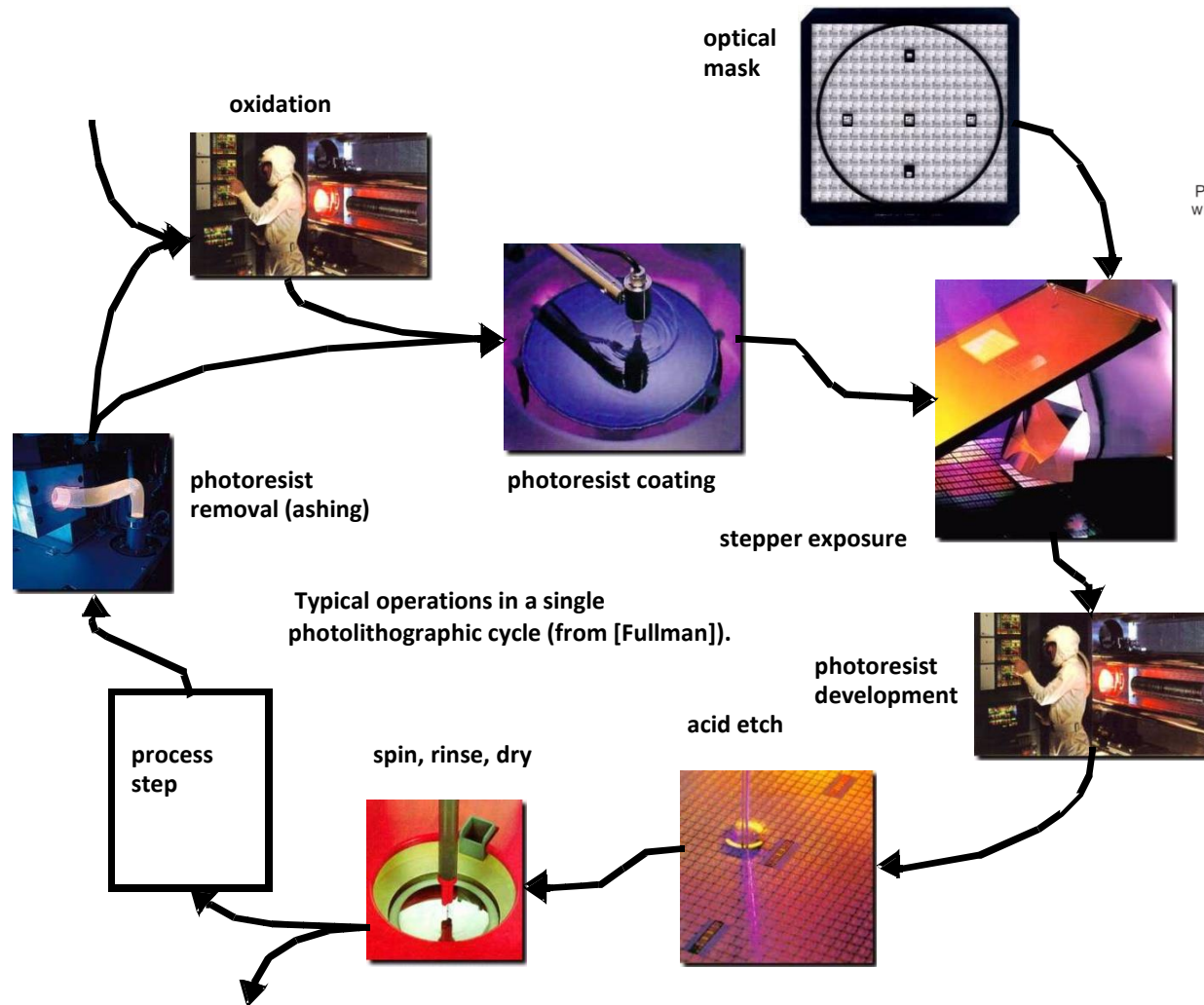
$2 \times 10^{21}$  impurities/m<sup>3</sup>



Digital IC Design



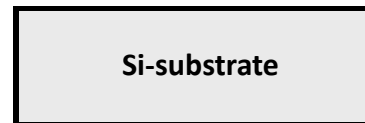
# Photo-Lithographic Process



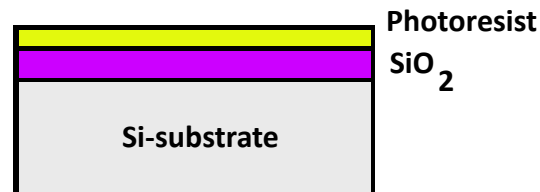




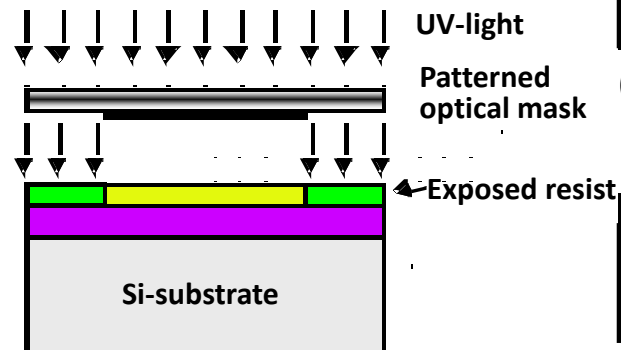
# Patterning of SiO<sub>2</sub>



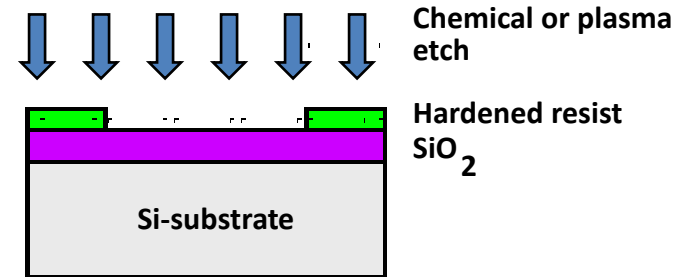
(a) Silicon base material



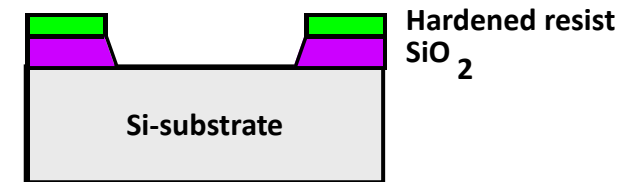
(b) After oxidation and deposition of negative photoresist



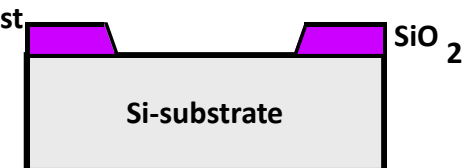
(c) Stepper exposure



(d) After development and etching of resist, chemical or plasma etch of SiO<sub>2</sub>



(e) After etching

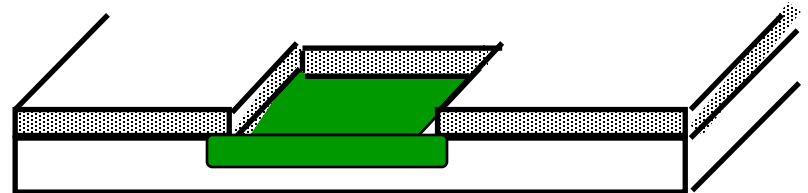
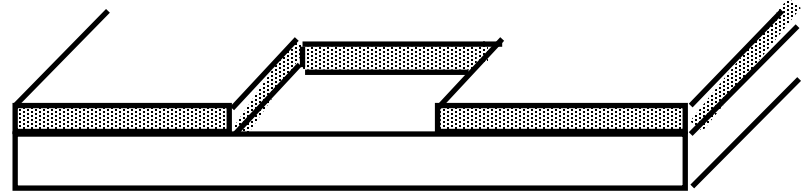


(f) Final result after removal of resist



# Diffusion and Ion Implantation

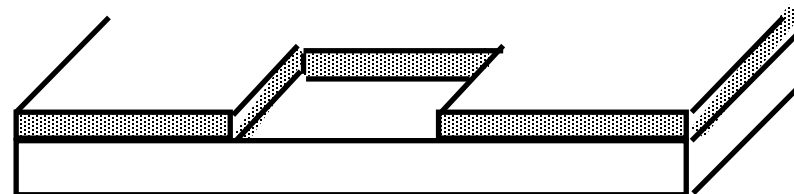
1. Area to be doped is exposed (photolithography)
2. Diffusion or Ion implantation





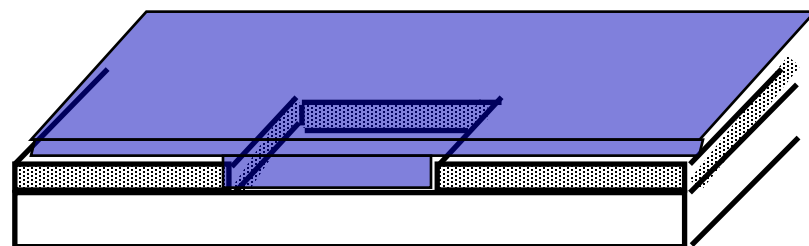
# Deposition (淀积) and Etching

1. Pattern masking  
(photolithography)



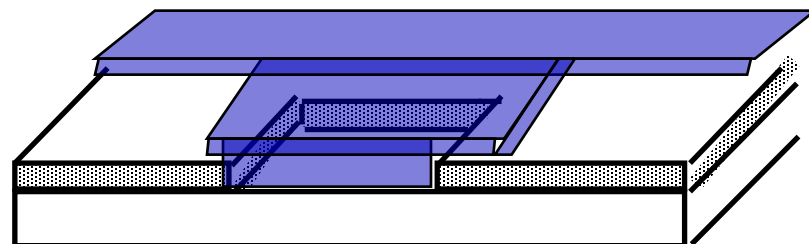
2. Deposit material over  
entire wafer

CVD ( $\text{Si}_3\text{N}_4$ )  
chemical deposition  
(polysilicon)  
sputtering (Al)



3. Etch away unwanted  
material

wet etching  
dry (plasma) etching





# Planarization: Polishing the Wafers





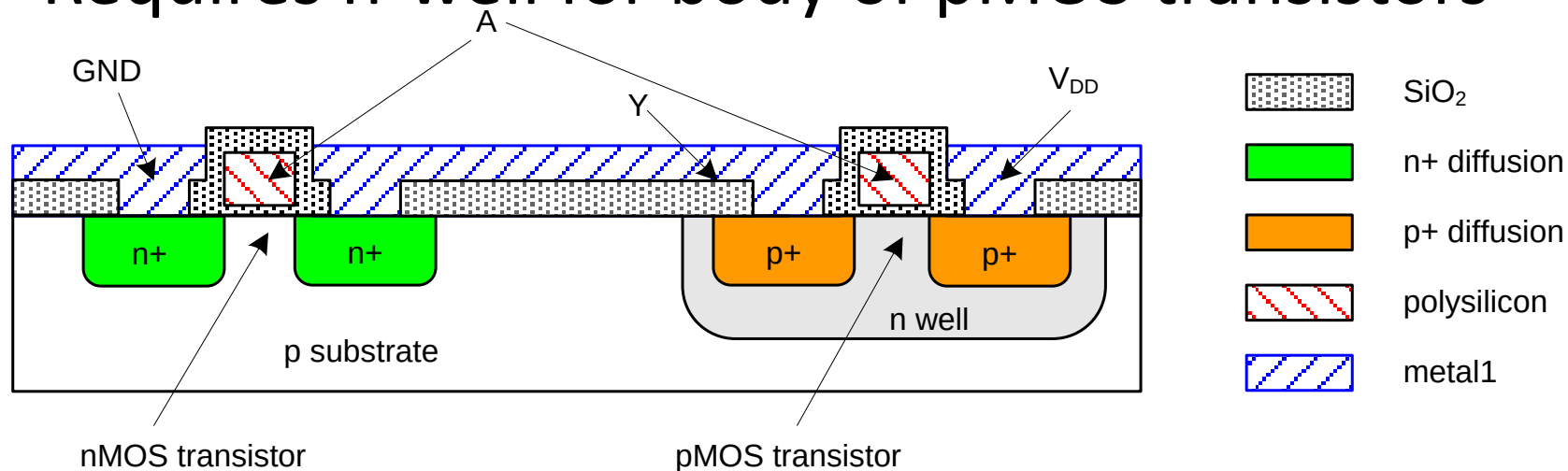
# CMOS Fabrication

- CMOS transistors are fabricated on silicon wafer
- Lithography process similar to printing press
- On each step, different materials are deposited or etched
- Easiest to understand by viewing both top and cross-section of wafer in a simplified manufacturing process



# Inverter Cross-section

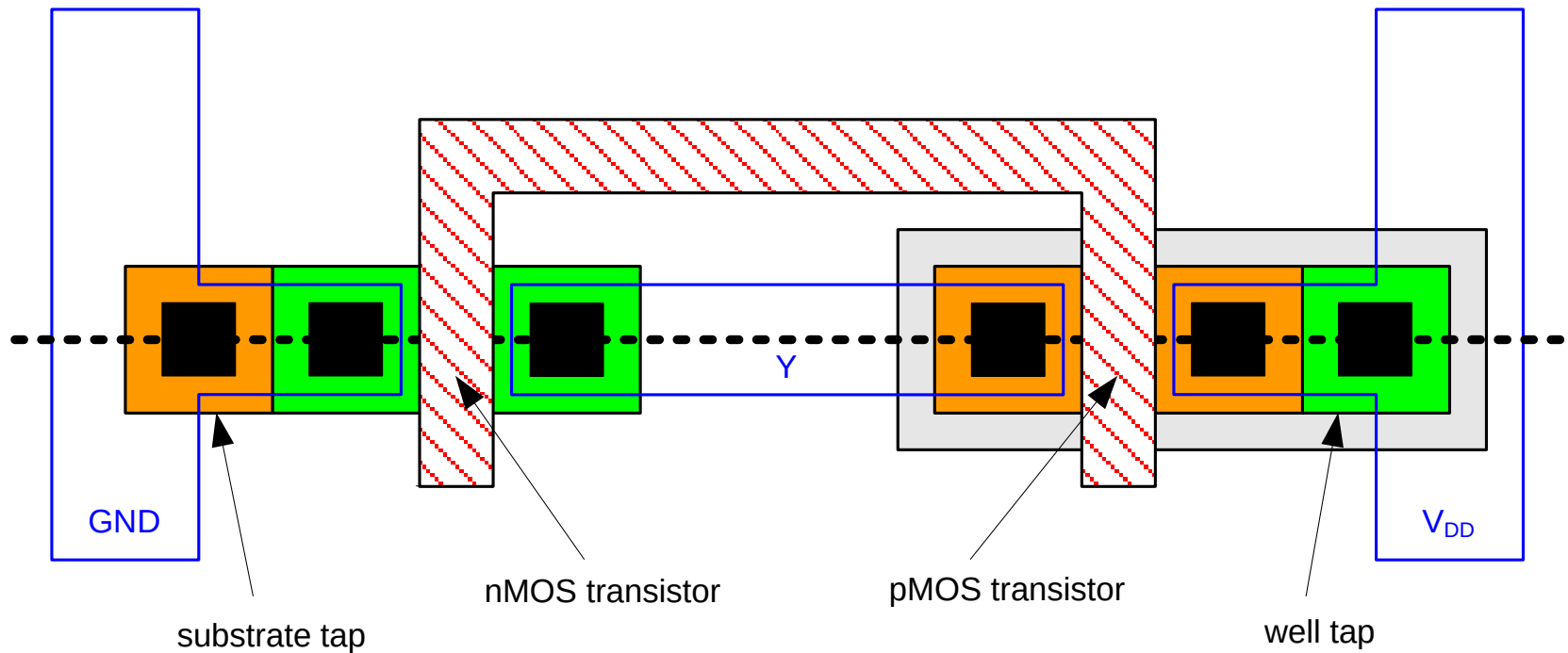
- Typically use p-type substrate for nMOS transistors
- Requires n-well for body of pMOS transistors





# Inverter Mask Set

- Transistors and wires are defined by *masks*
- Cross-section taken along dashed line

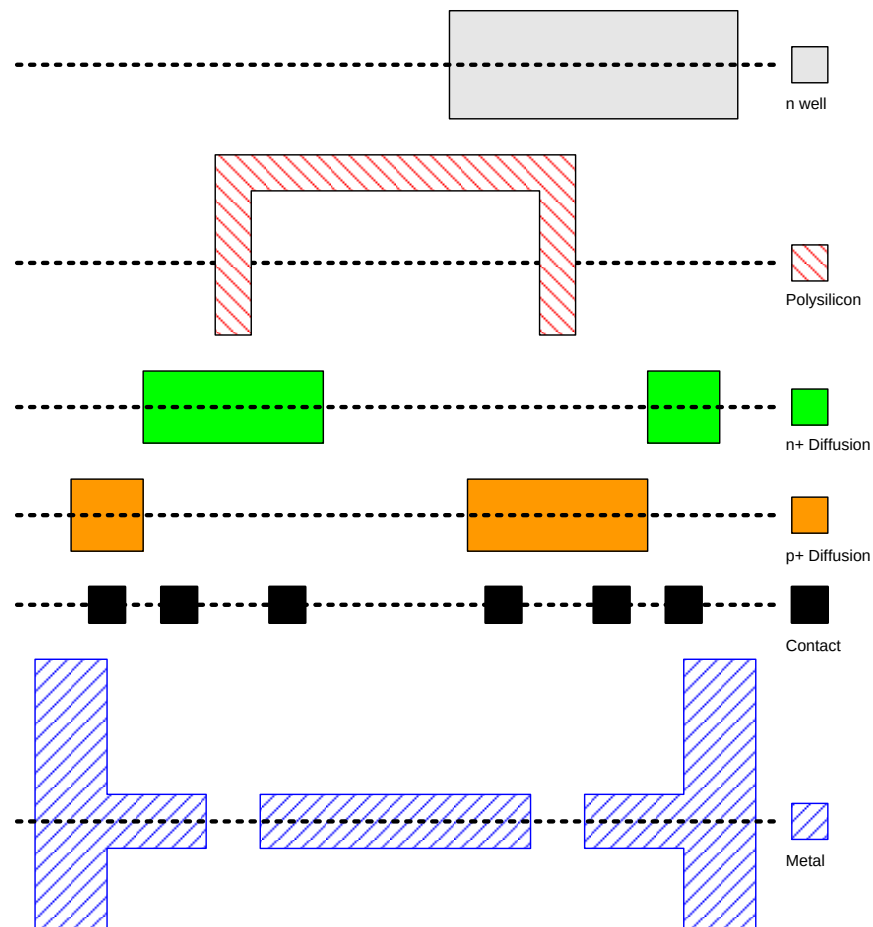






# Detailed Mask Views

- Six masks
  - n-well
  - Polysilicon
  - n+ diffusion
  - p+ diffusion
  - Contact
  - Metal





# Fabrication Steps

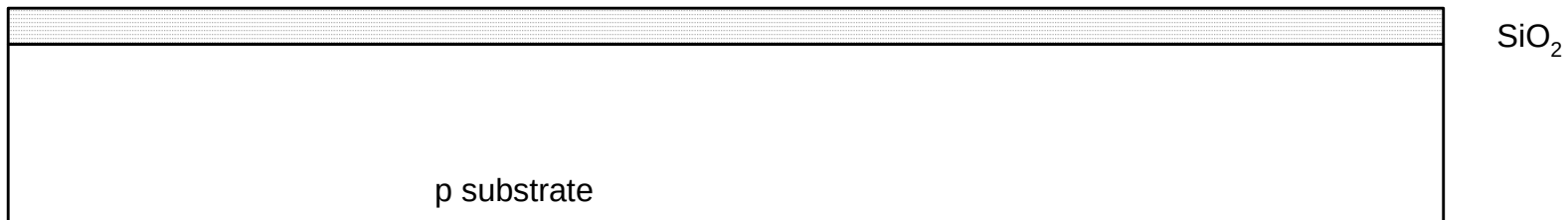
- Start with blank wafer
- Build inverter from the bottom up
- First step will be to form the n-well
  - Cover wafer with protective layer of  $\text{SiO}_2$  (oxide)
  - Remove layer where n-well should be built
  - Implant or diffuse n dopants into exposed wafer
  - Strip off  $\text{SiO}_2$

p substrate



# Oxidation

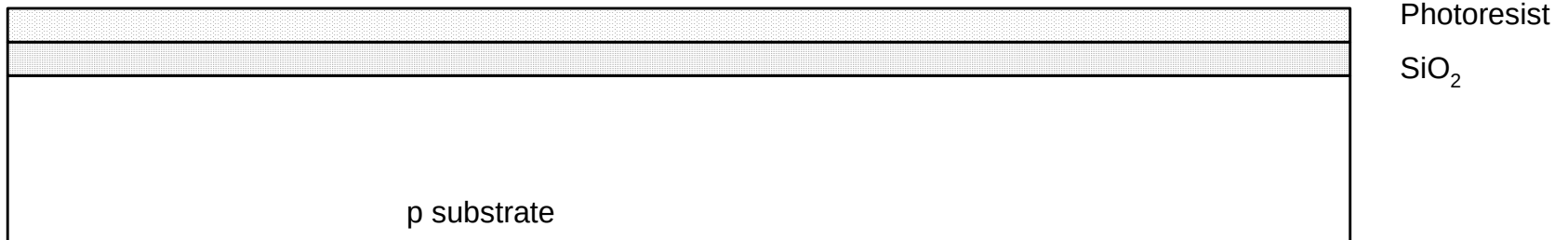
- Grow  $\text{SiO}_2$  on top of Si wafer
  - 900 – 1200 C with  $\text{H}_2\text{O}$  or  $\text{O}_2$  in oxidation furnace





# Photoresist

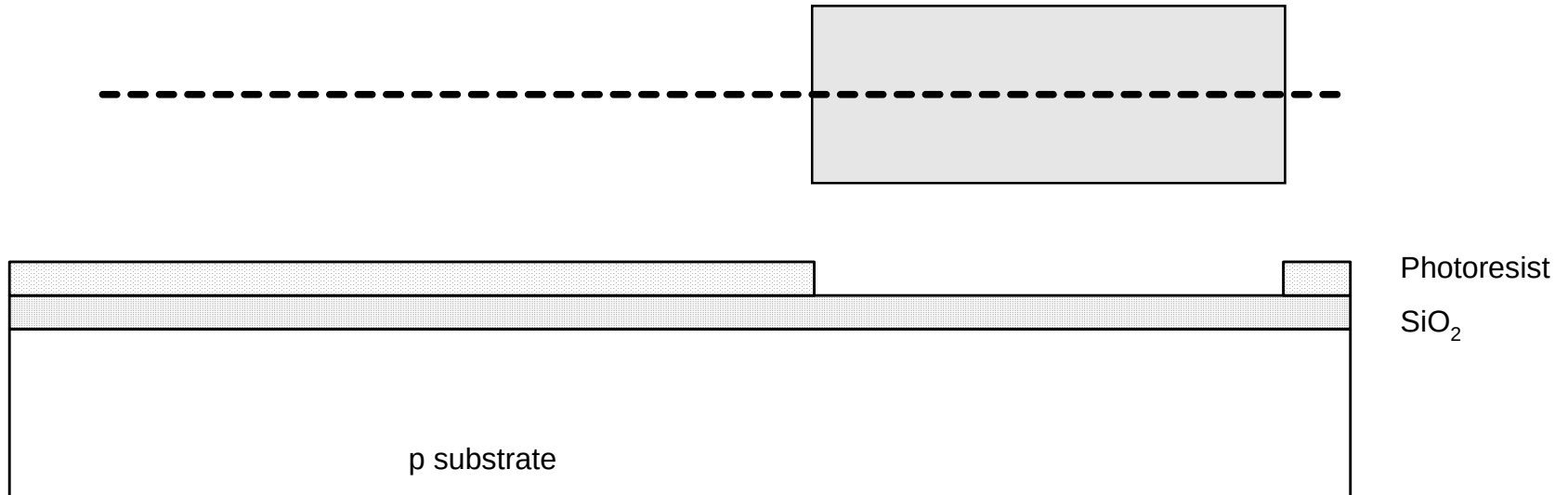
- Spin on photoresist
  - Photoresist is a light-sensitive organic polymer
  - Softens where exposed to light





# Lithography

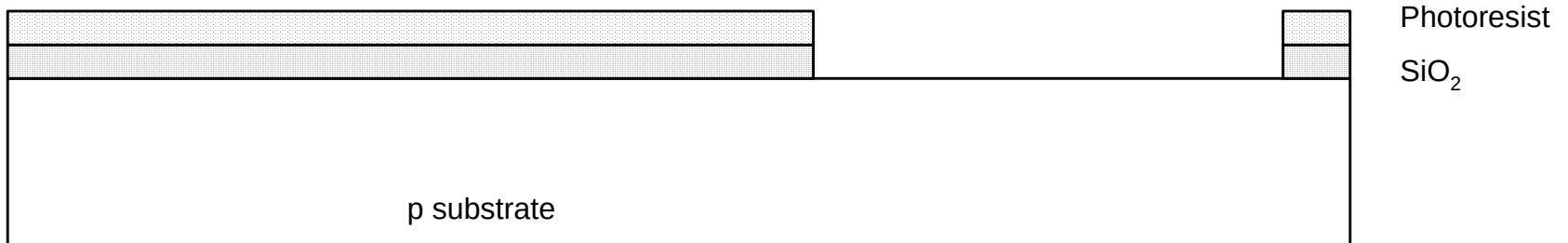
- Expose photoresist through n-well mask
- Strip off exposed photoresist





# Etch

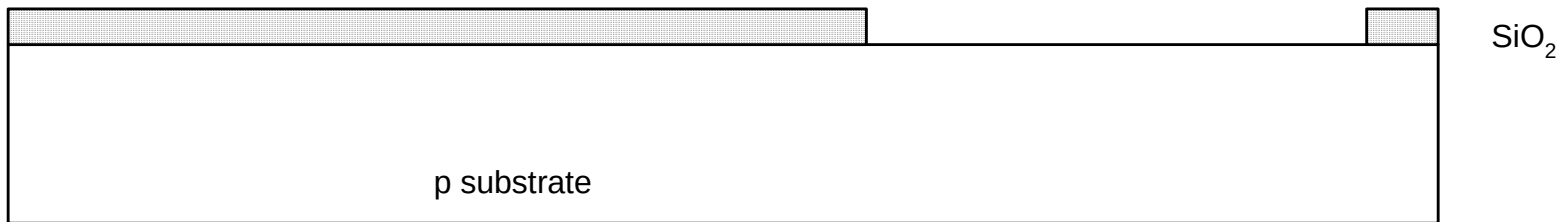
- Etch oxide with hydrofluoric acid (HF)
- Only attacks oxide where resist has been exposed





# Strip Photoresist

- Strip off remaining photoresist
  - Use mixture of acids called piranah etch
- Necessary so resist doesn't melt in next step

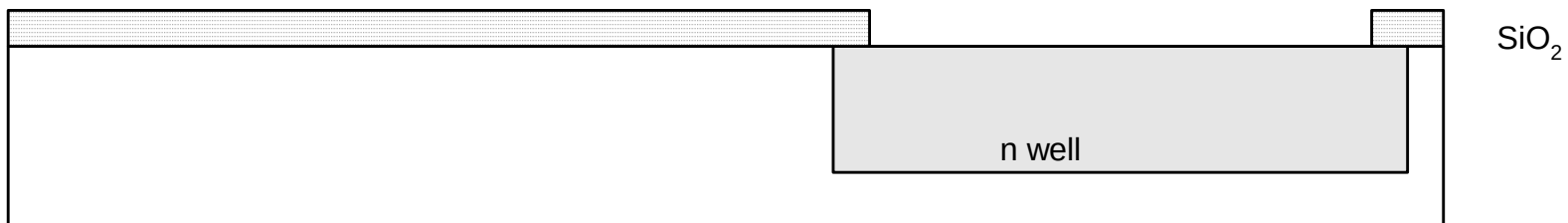






# n-well

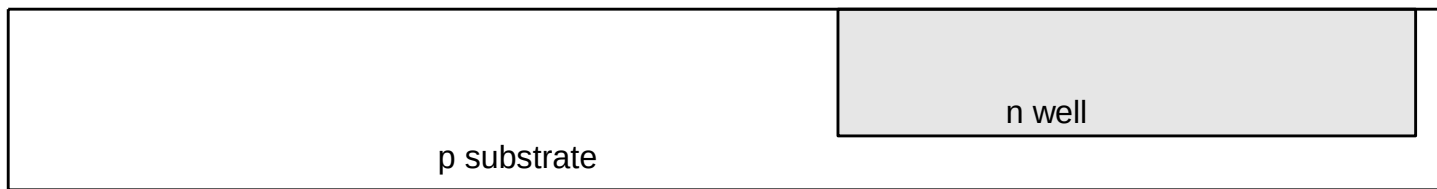
- n-well is formed with diffusion or ion implantation
- Diffusion
  - Place wafer in furnace with arsenic gas
  - Heat until As atoms diffuse into exposed Si
- Ion Implantation
  - Blast wafer with beam of As ions
  - Ions blocked by  $\text{SiO}_2$ , only enter exposed Si





# Strip Oxide

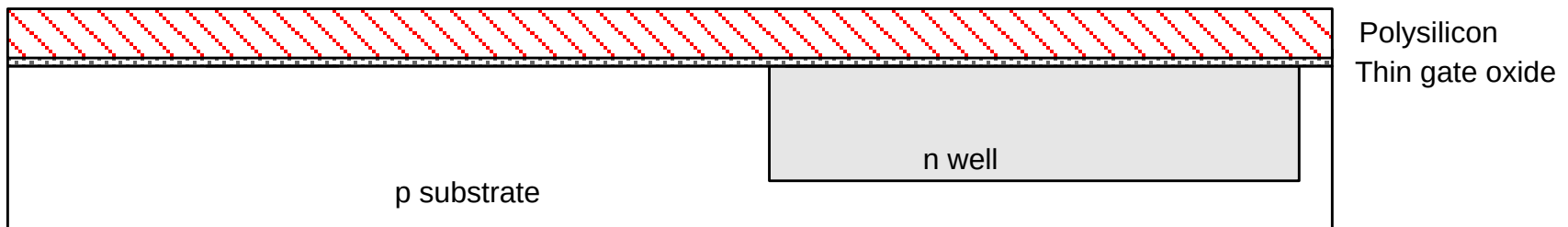
- Strip off the remaining oxide using HF
- Back to bare wafer with n-well
- Subsequent steps involve similar series of steps





# Polysilicon

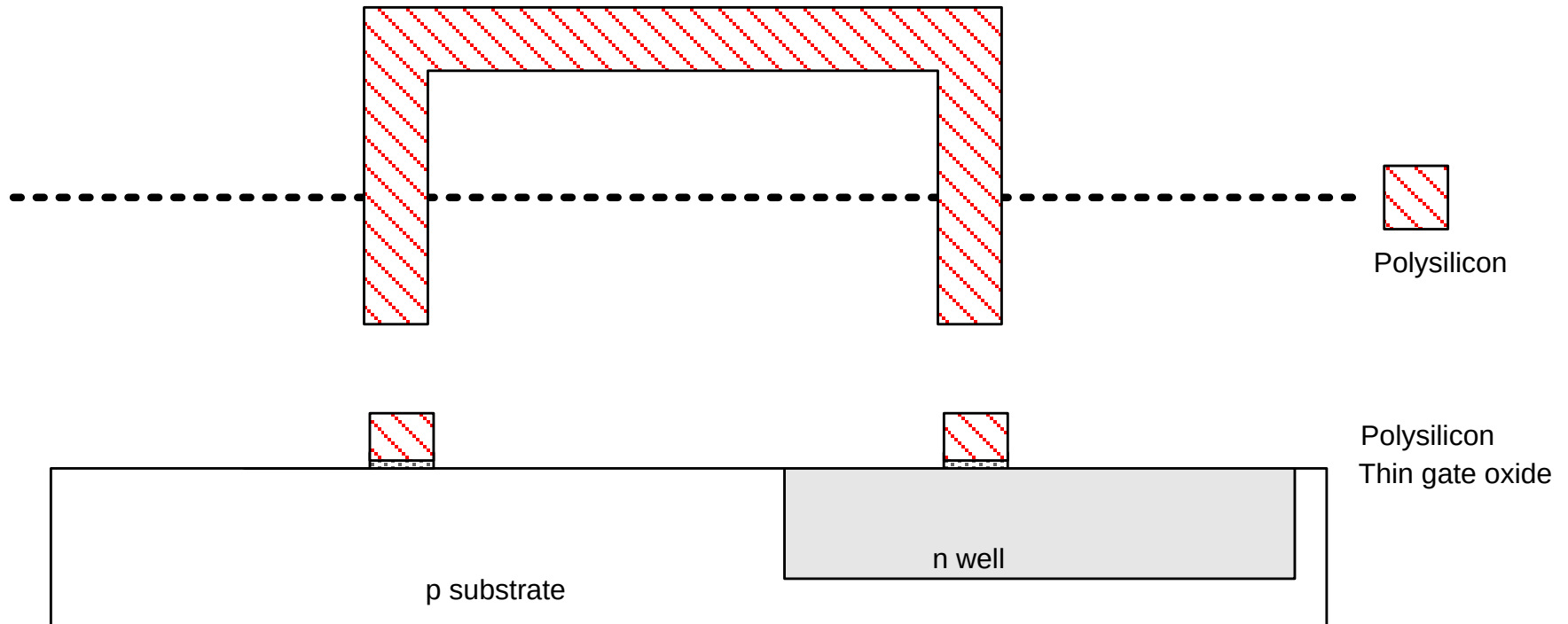
- Deposit very thin layer of gate oxide
  - $< 20 \text{ \AA}$  (6-7 atomic layers)
- Chemical Vapor Deposition (CVD) of silicon layer
  - Place wafer in furnace with Silane gas ( $\text{SiH}_4$ )
  - Forms many small crystals called polysilicon
  - Heavily doped to be good conductor





# Polysilicon Patterning

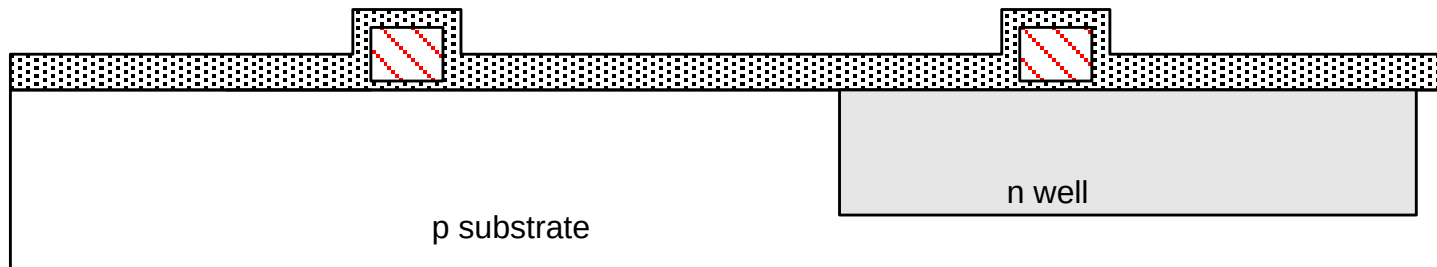
- Use same lithography process to pattern polysilicon





# Self-Aligned Process

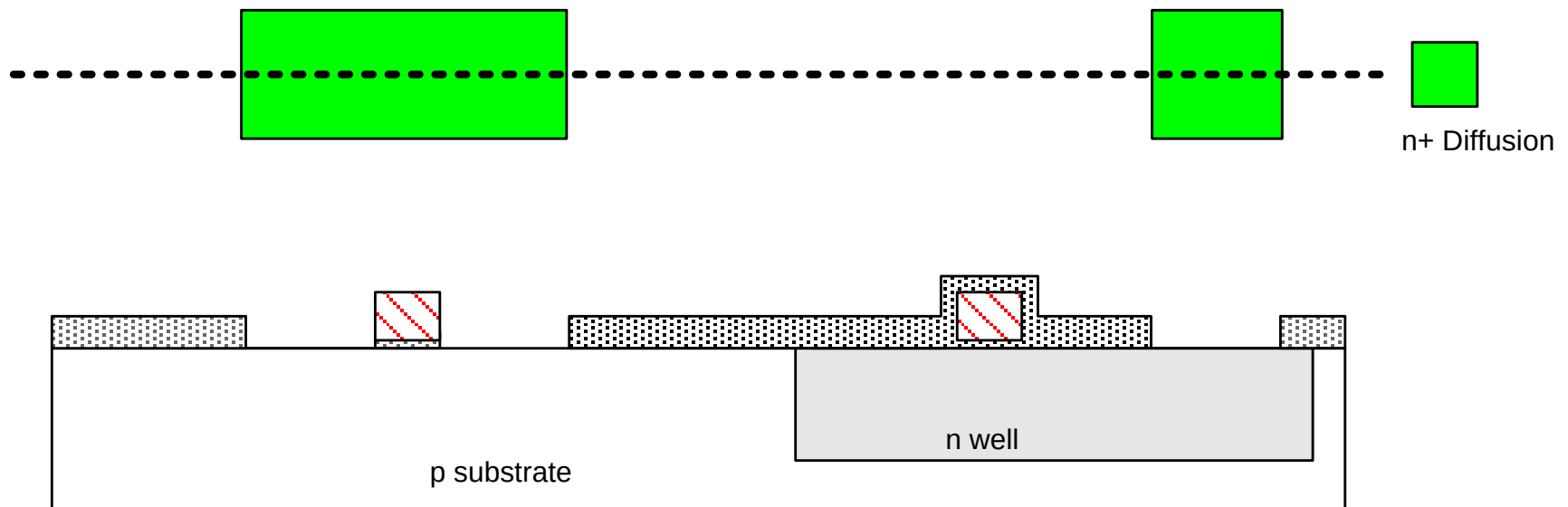
- Use oxide and masking to expose where n+ dopants should be diffused or implanted
- N-diffusion forms nMOS source, drain, and n-well contact





# N-diffusion

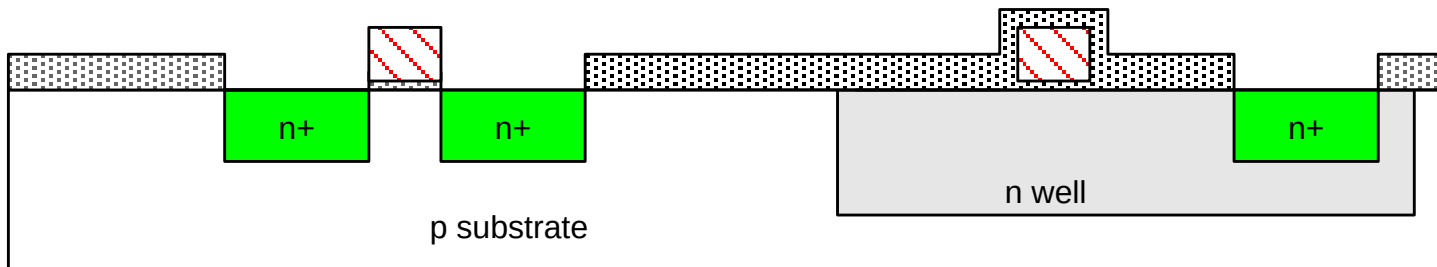
- Pattern oxide and form n+ regions
- *Self-aligned process* where gate blocks diffusion
- Polysilicon is better than metal for self-aligned gates because it doesn't melt during later processing





## N-diffusion cont.

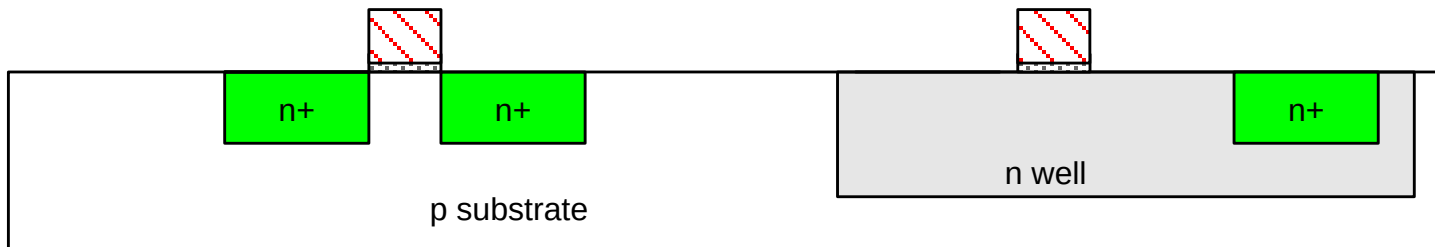
- Historically dopants were diffused
- Usually ion implantation today
- But regions are still called diffusion





## N-diffusion cont.

- Strip off oxide to complete patterning step

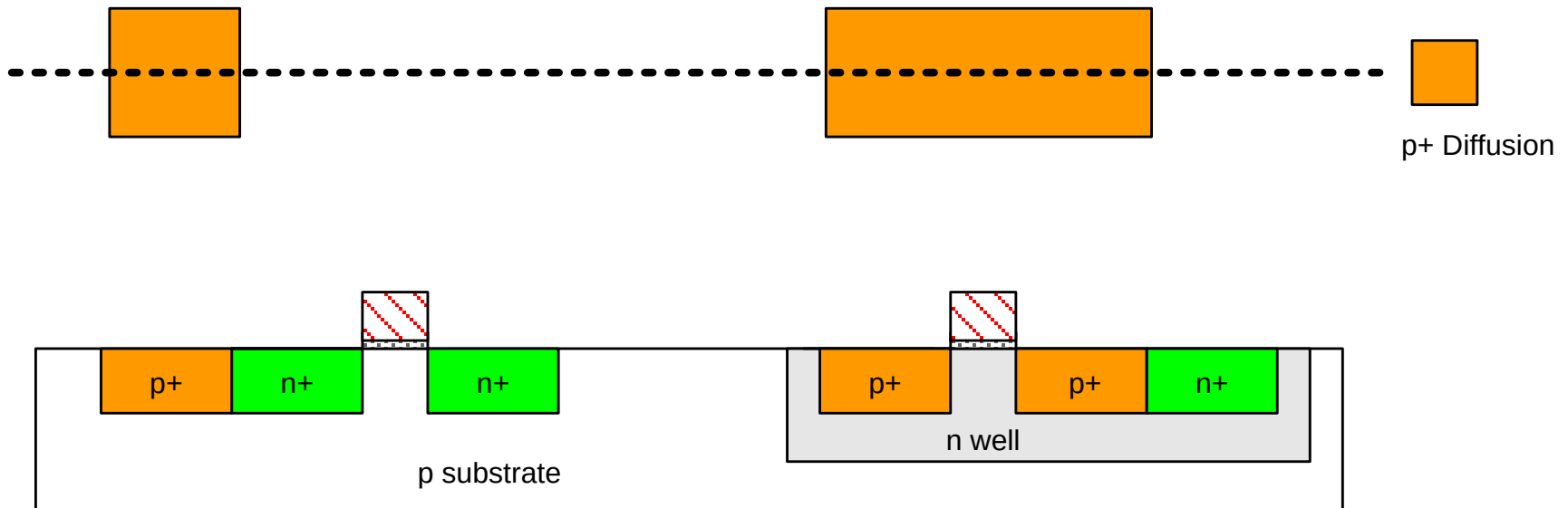






# P-Diffusion

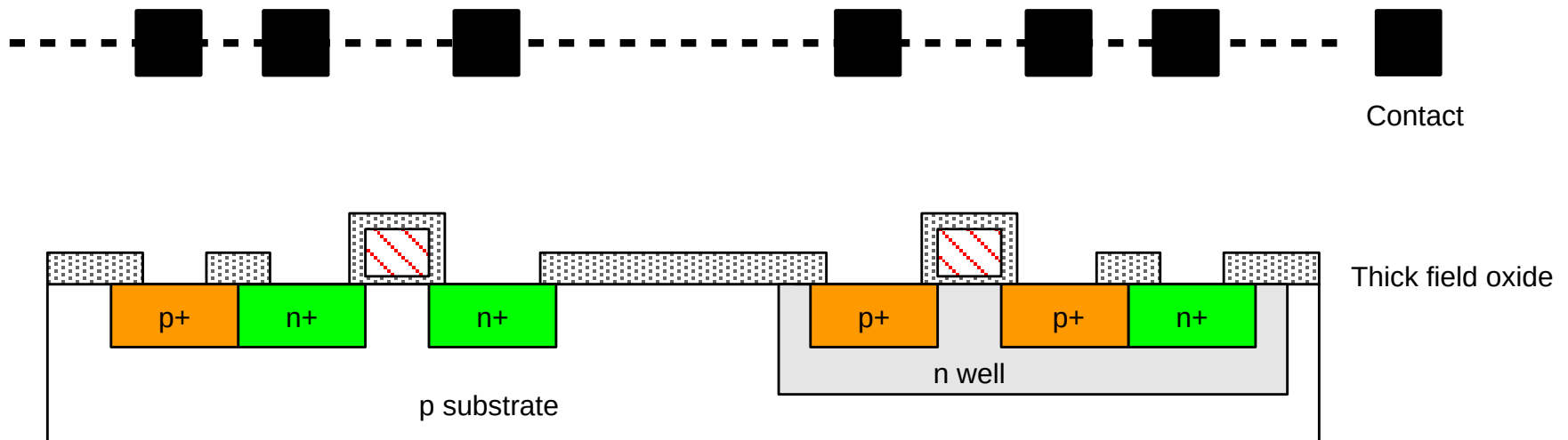
- Similar set of steps form p+ diffusion regions for pMOS source and drain and substrate contact





# Contacts

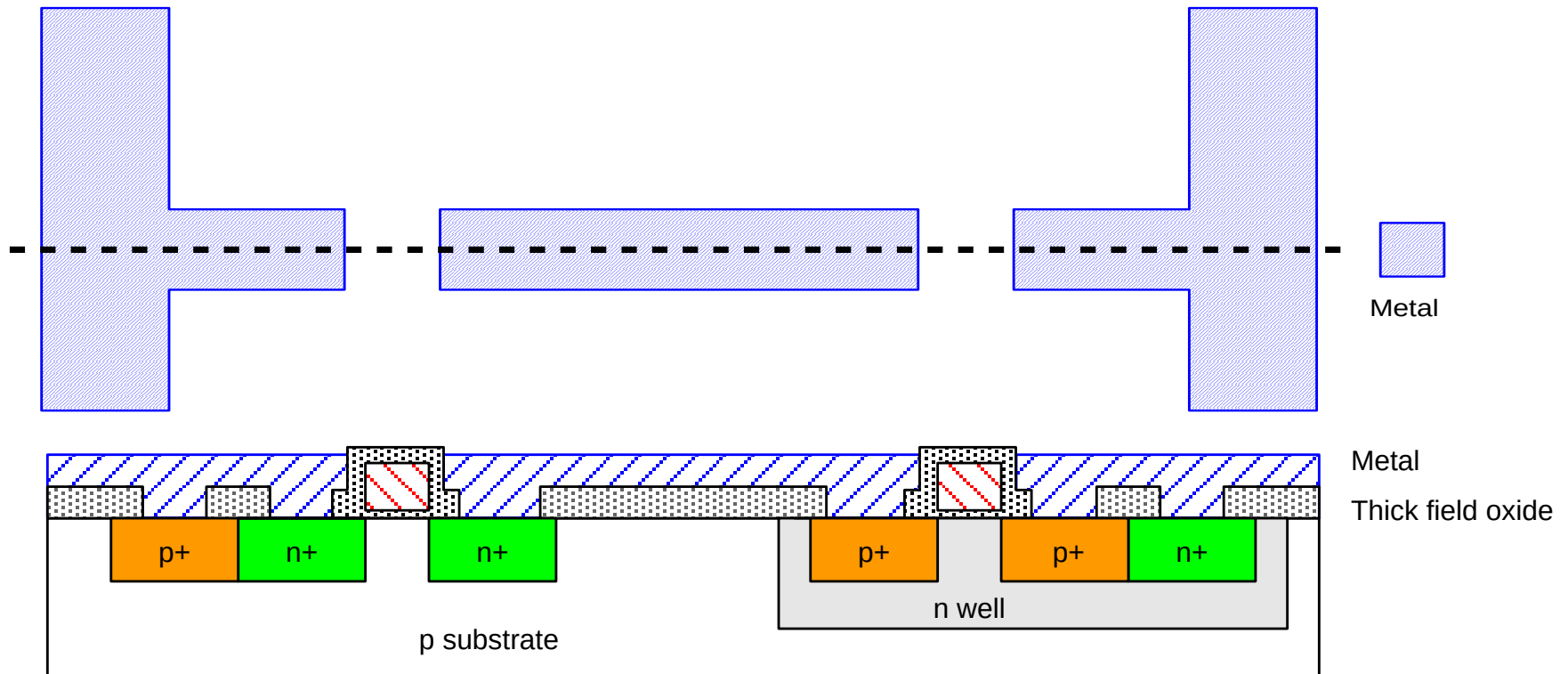
- Now we need to wire together the devices
- Cover chip with thick field oxide
- Etch oxide where contact cuts are needed





# Metalization

- Sputter on aluminum over whole wafer
- Pattern to remove excess metal, leaving wires



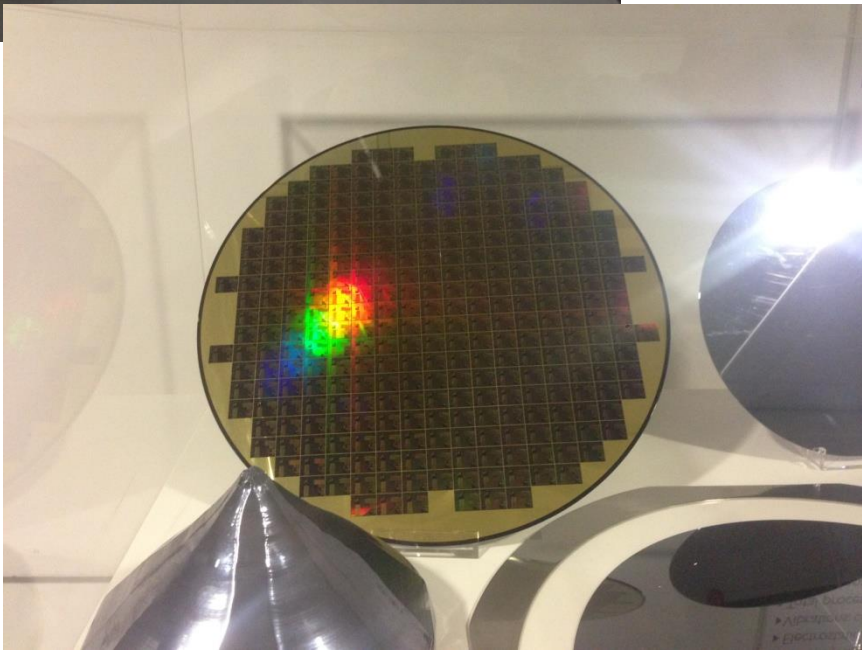


# Fabrication

- Chips are built in huge factories called fabs
- Contain clean rooms as large as football fields



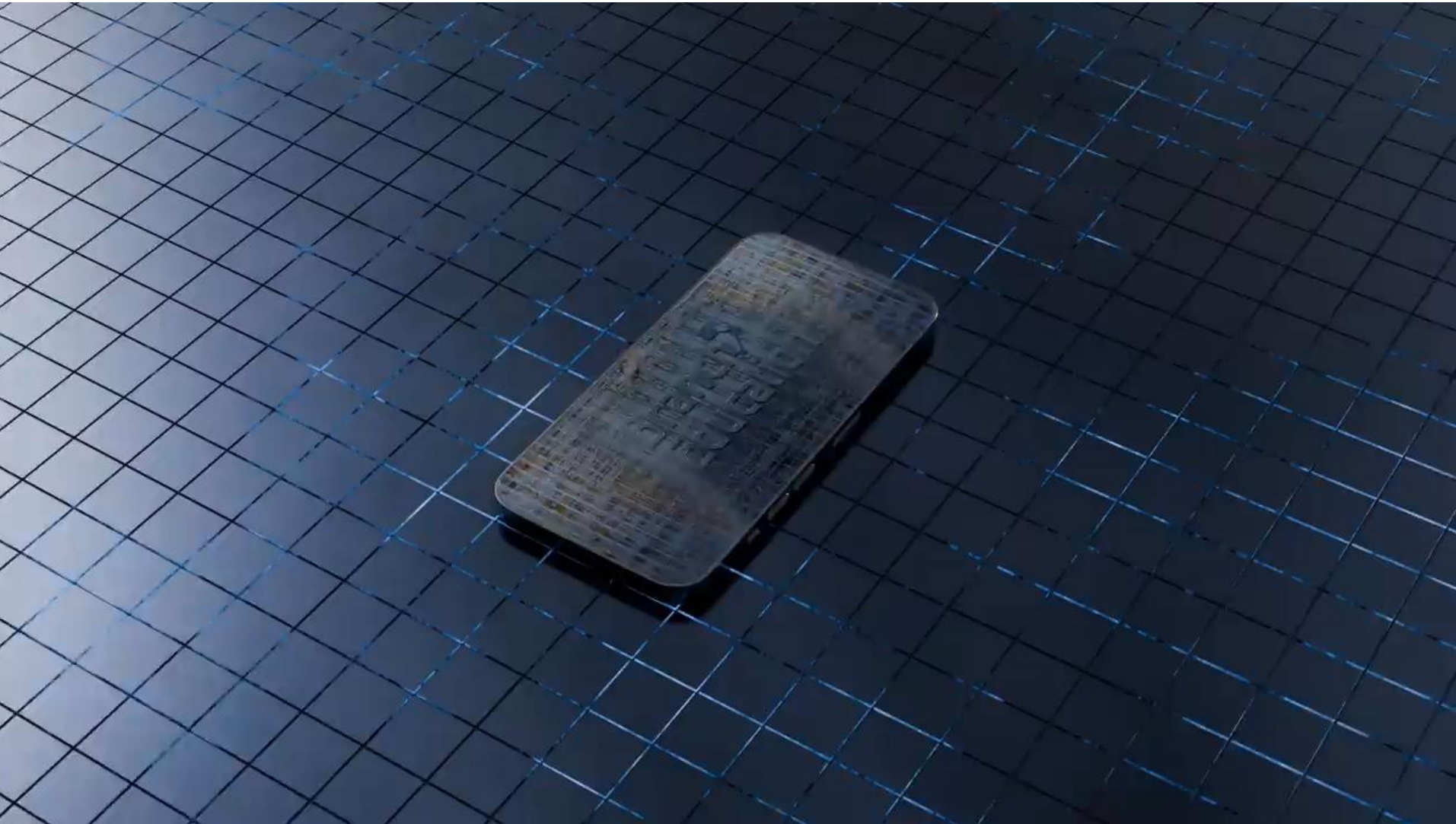
Courtesy of International  
Business Machines Corporation.  
Unauthorized use not permitted.







# How are Microchips Made



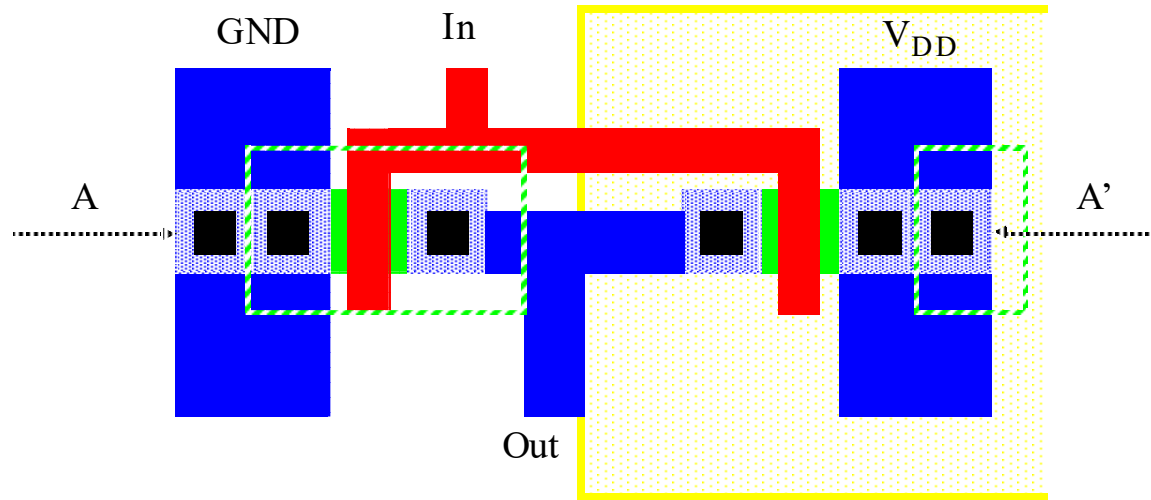


# Layout

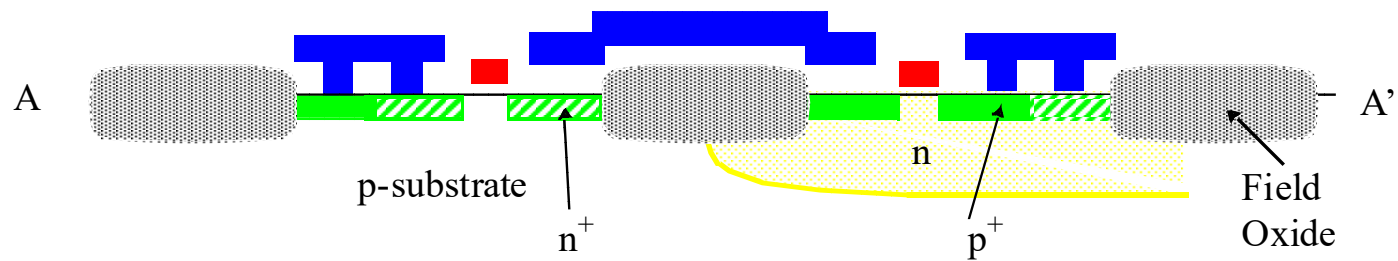
- Chips are specified with set of masks
- Minimum dimensions of masks determine transistor size (and hence speed, cost, and power)
- Feature size  $f$  = distance between source and drain
  - Set by minimum width of polysilicon
- Feature size improves 30% every 3 years or so



# CMOS Inverter Layout



(a) Layout



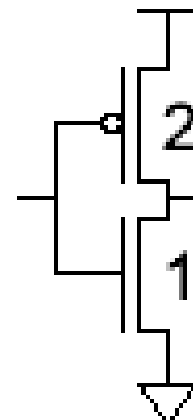
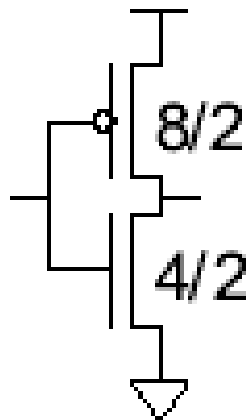
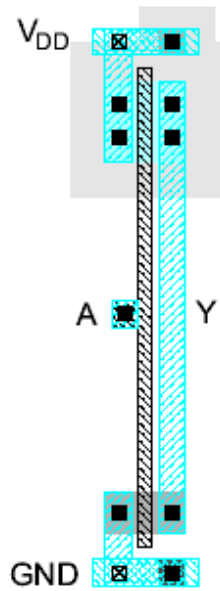
(b) Cross-Section along A-A'





# Inverter Layout

- Transistor dimensions specified as Width / Length
  - Minimum size is  $4\lambda / 2\lambda$ , sometimes called 1 unit
  - In  $f = 0.6 \mu\text{m}$  process, this is  $1.2 \mu\text{m}$  wide,  $0.6 \mu\text{m}$  long





# Design Rules

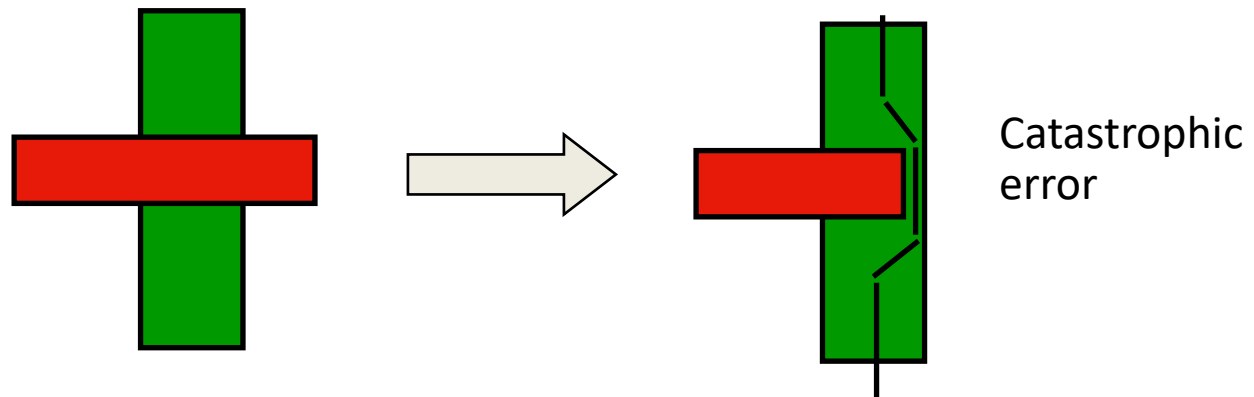
- Interface between designer and process engineer
- Guidelines for constructing process masks
- Unit dimension: Minimum line width
  - scalable design rules: lambda parameter  $\lambda$
  - absolute dimensions (micron rules)
- Normalize for feature size when describing design rules
- Express rules in terms of  $\lambda = f/2$ 
  - E.g.  $\lambda = 0.3 \mu\text{m}$  in  $0.6 \mu\text{m}$  process
- Rules constructed to ensure that design works even when small fab errors (within some tolerance) occur
- A complete set includes
  - set of layers
  - intra-layer: relations between objects in the same layer
  - inter-layer: relations between objects on different layers



## why

- Minimum dimensions (e.g., widths) of objects on each layer to maintain that object after fab.
- Minimum spaces between objects (that are *not* related) on the same layer to ensure they will not short after fab.

Transistors



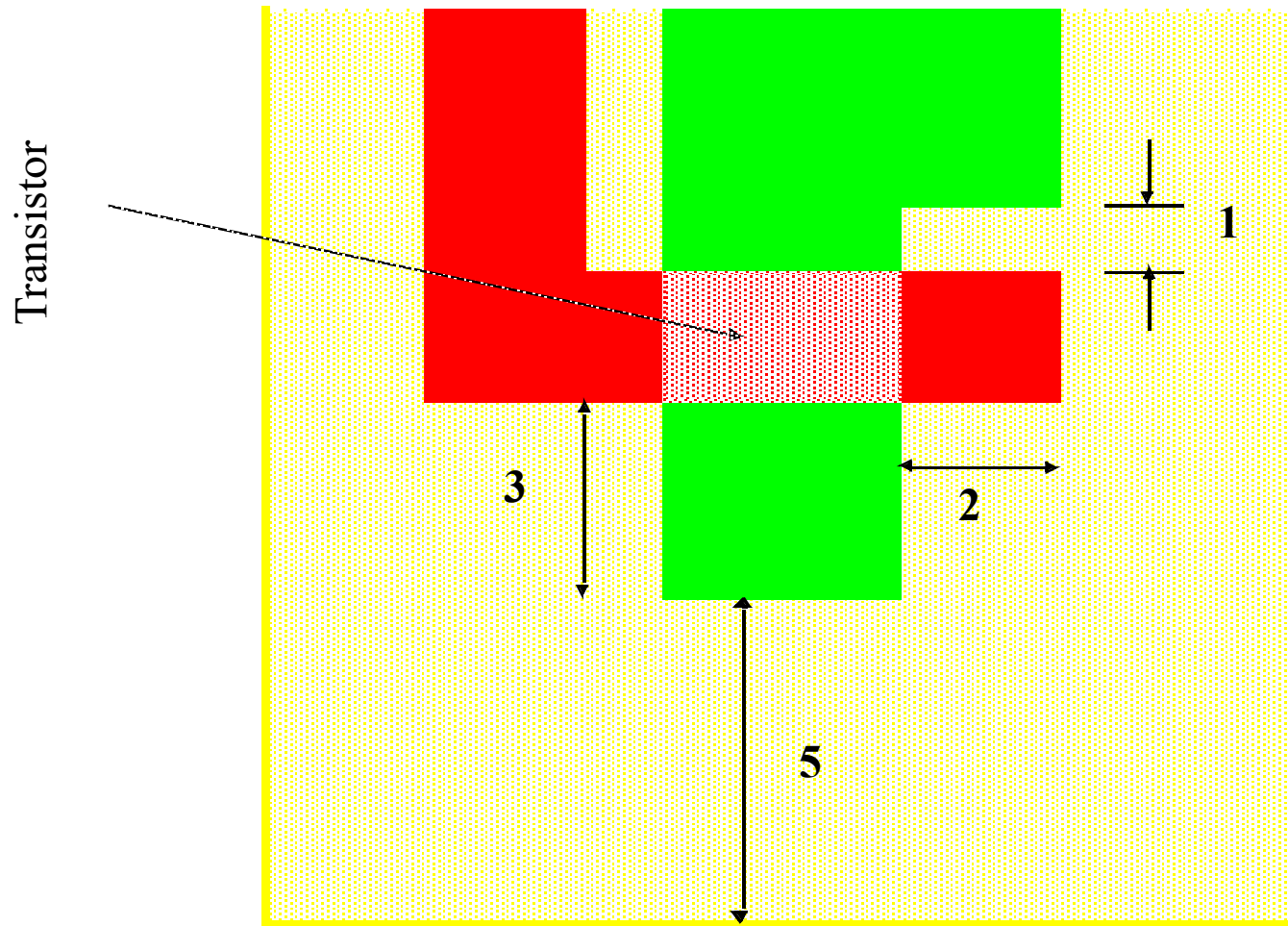


# CMOS Process Layers

| Layer                | Color   | Representation                                                                        |
|----------------------|---------|---------------------------------------------------------------------------------------|
| Well (p,n)           | Yellow  |    |
| Active Area (n+,p+)  | Green   |    |
| Select (p+,n+)       | Green   |    |
| Polysilicon          | Red     |    |
| Metal1               | Blue    |    |
| Metal2               | Magenta |    |
| Contact To Poly      | Black   |   |
| Contact To Diffusion | Black   |  |
| Via                  | Black   |  |



# Transistor Layout





- 
- The diagram illustrates the layout rules for a 1.5-μm CMOS technology, showing various components and their dimensions in terms of  $\lambda$ .
- Metal1:** Two vertical bars representing Metal1. The width of each bar is  $4\lambda$ . The spacing between the bars is  $4\lambda$ .
  - Metal2:** Two vertical bars representing Metal2. The width of each bar is  $4\lambda$ . The spacing between the bars is  $4\lambda$ .
  - Diffusion:** Two vertical bars representing Diffusion. The width of each bar is  $4\lambda$ . The spacing between the bars is  $4\lambda$ .
  - Polysilicon:** Two vertical bars representing Polysilicon. The width of each bar is  $2\lambda$ . The spacing between the bars is  $3\lambda$ .
  - n-well:** A large rectangular area representing the n-well. The width is  $6\lambda$  and the height is  $6\lambda$ . The spacing between the n-well and the Metal1 bars is  $2\lambda$ .
  - Metal1-Diffusion Contact:** A small square representing a contact. The width is  $1\lambda$  and the height is  $2\lambda$ .
  - Metal1-Polysilicon Contact:** A small square representing a contact. The width is  $1\lambda$  and the height is  $2\lambda$ .
  - Metal1-Metal2 Vias:** A small square representing a via. The width is  $1\lambda$  and the height is  $2\lambda$ . The spacing between the vias is  $3\lambda$ .

# 0.18um Mixed-Mode/RFCMOS 1.8V/3.3V 1P6M Metal Metal Capacitor Process Topological Layout Rule

(Ver. 2.4\_P1)

Single Poly 6 Metal (1P6M), Dual Voltage (1.8V & 3.3V)  
P-Sub, TwiN\_WELL/T\_WELL CMOS Process  
@ 1.8V/3.3Volt Operation

SPEC NO:G-03-MIXED\_MODE/RFCMOS18-1.8V/3.3V-1P6M-MMC-TLR

Ver:2.4 Phase:1 Approved Date:06/22/2004

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## 1.20 METAL1 Layer (No. P60) :

(unit : um)

|      |                                  |      |
|------|----------------------------------|------|
| 20A. | Minimum METAL1 width             | 0.24 |
| 20B. | Minimum METAL1 to METAL1 spacing |      |
| a.   | METAL1 width < 10 um             | 0.24 |
| b.   | METAL1 width ≥ 10 um             | 0.28 |

|      |                                  |     |
|------|----------------------------------|-----|
| 20C. | Maximum METAL1 to METAL1 spacing | 500 |
|------|----------------------------------|-----|

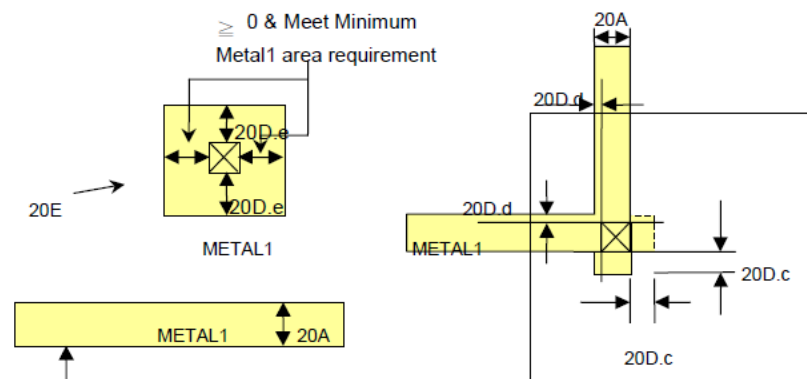
|      |                                                                                                                        |      |
|------|------------------------------------------------------------------------------------------------------------------------|------|
| 20D. | Minimum METAL1 overlap CONTACT                                                                                         |      |
| a.   | METAL1 line overlap CONTACT                                                                                            | 0    |
| b.   | METAL1 line end overlap CONTACT                                                                                        | 0.08 |
| c.   | METAL1 outer corner overlap CONTACT<br>(for two adjacent sides, at least one side should be treated as metal line end) | 0.08 |
| d.   | METAL1 inner corner overlap CONTACT                                                                                    | 0    |
| e.   | METAL1 extension over CONTACT on opposite sides                                                                        | 0.08 |

|      |                     |                        |
|------|---------------------|------------------------|
| 20E. | Minimum METAL1 area | 0.1764 um <sup>2</sup> |
|------|---------------------|------------------------|

20F. The METAL1 coverage must be larger than 30% of the entire chip area.

20G. Minimum METAL1 coverage for every millimeter square area is 25 %.

Note : It is strongly recommended that metal coverage should be evenly distributed in the whole chip; the more uniform, the better.

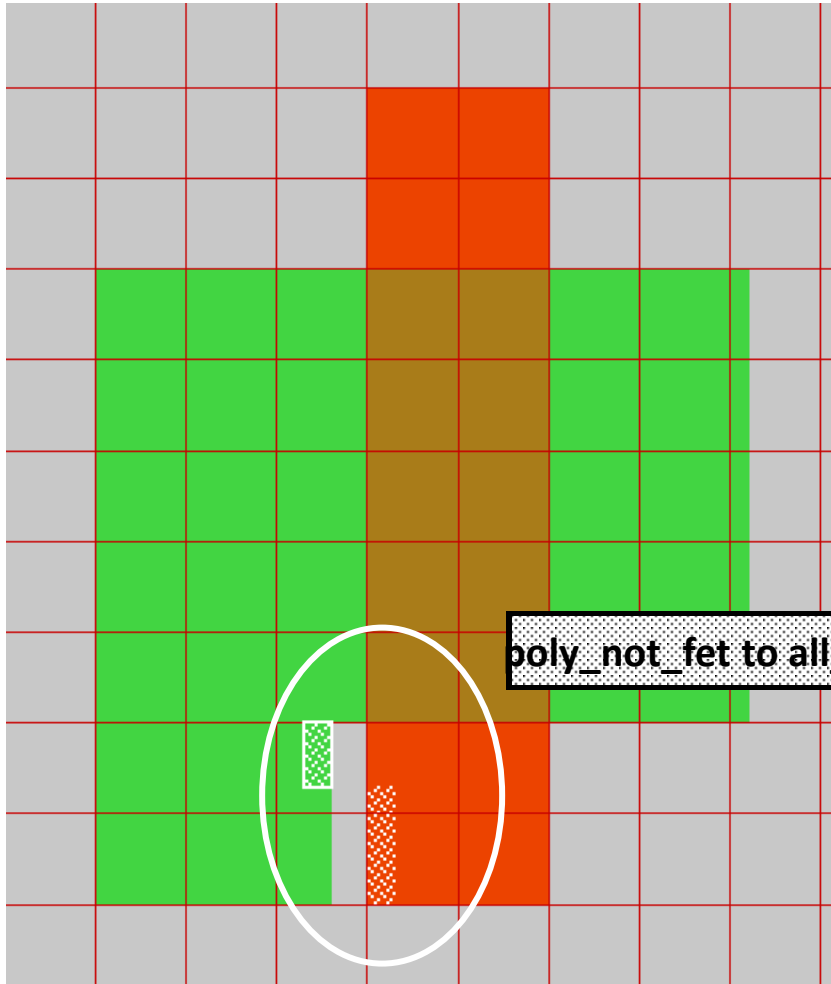


Index of design Rule

Design Rule of Metal 1



# Design Rule Checker



poly\_not\_fet to all\_diff minimum spacing = 0.14 um



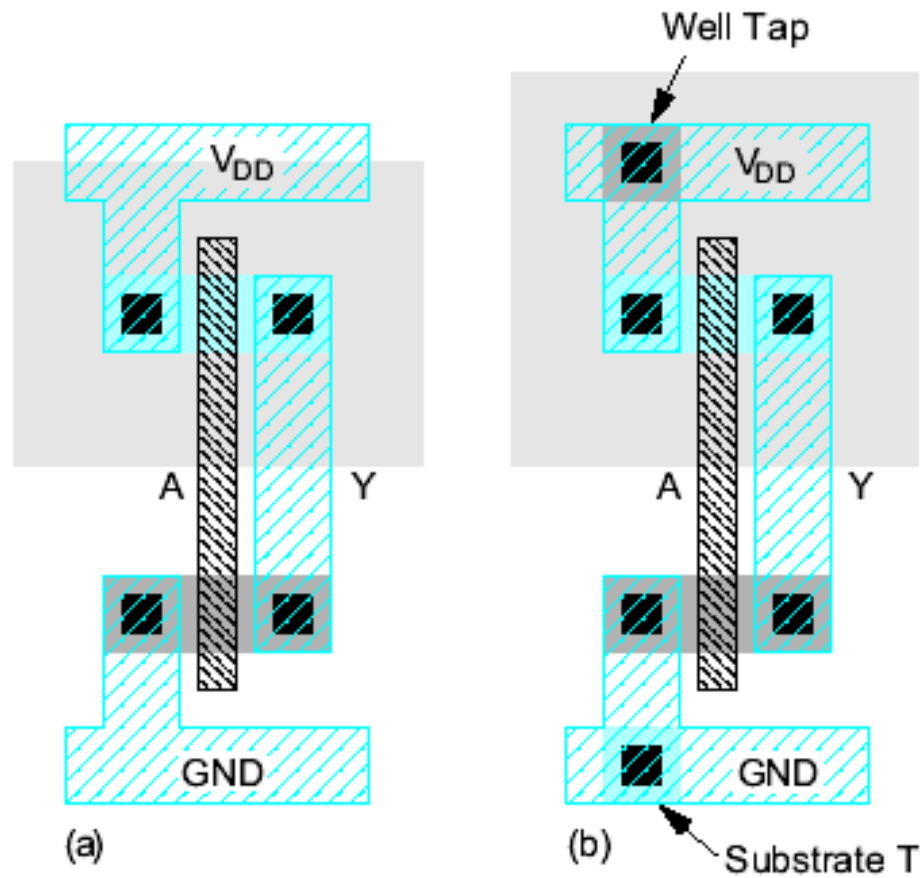


# Gate Layout

- Layout can be very time consuming
  - Design gates to fit together nicely
  - Build a library of standard cells
- Standard cell design methodology
  - $V_{DD}$  and GND should abut (standard height)
  - Adjacent gates should satisfy design rules
  - nMOS at bottom and pMOS at top
  - All gates include well and substrate contacts



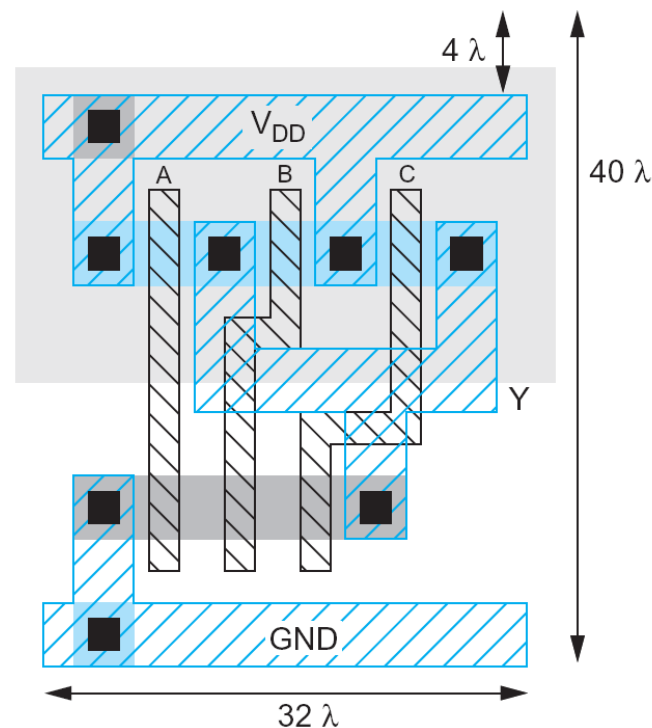
# Example: Inverter





## Example: NAND3

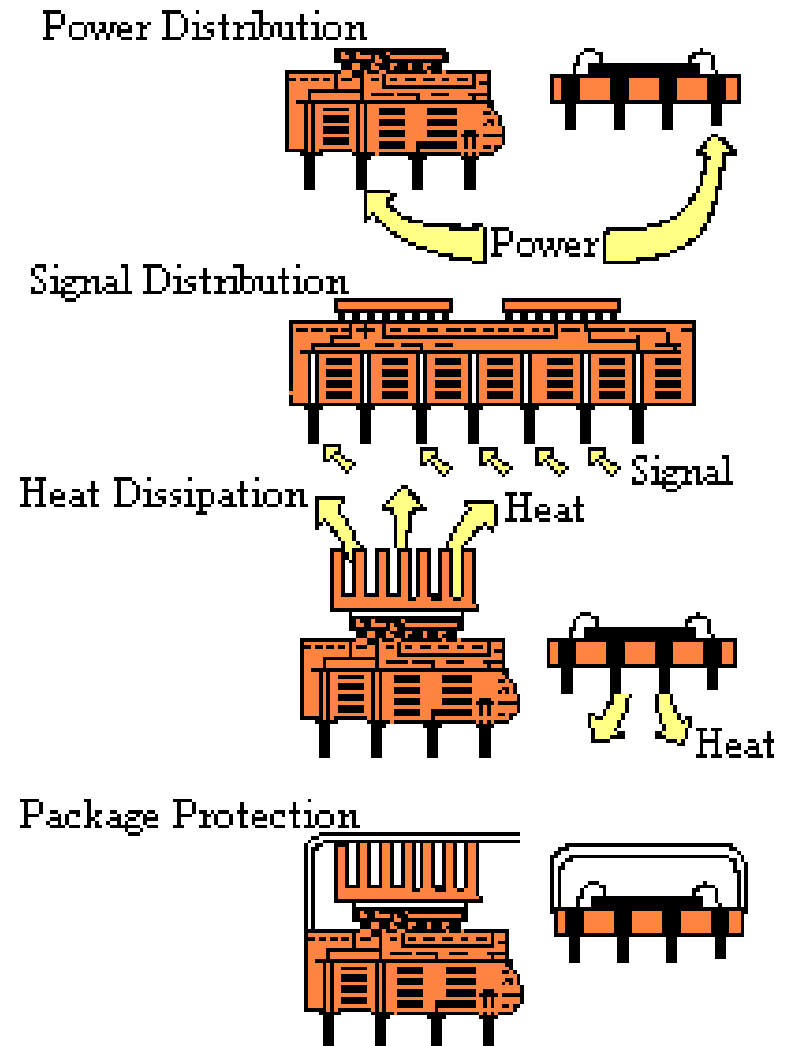
- Horizontal N-diffusion and p-diffusion strips
- Vertical polysilicon gates
- Metal1  $V_{DD}$  rail at top
- Metal1 GND rail at bottom
- $32\lambda$  by  $40\lambda$





# Packaging Requirements

- Electrical
  - Low parasitics
- Mechanical
  - Reliable and robust
- Thermal
  - Efficient heat removal
- Economical
  - Cheap
- Delay: HPCS 50%





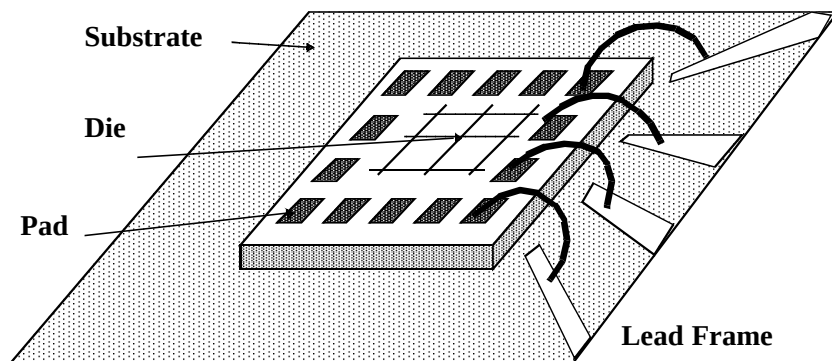
# Interconnect

- Interconnect level 1 die-to-package substrate
- Interconnect level 2 Package-Substrate-to-Board
- Die-to-Board



# Interconnect level 1 die-to-package substrate

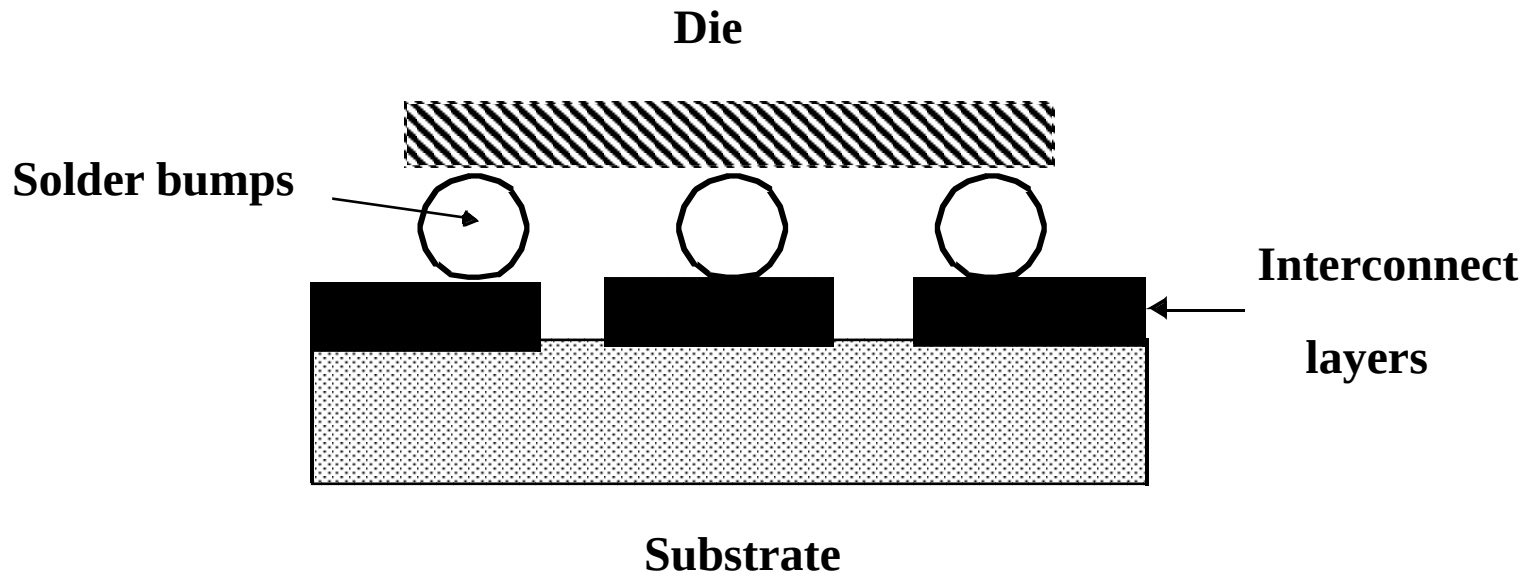
- Wire Bonding Techniques(导线压焊)





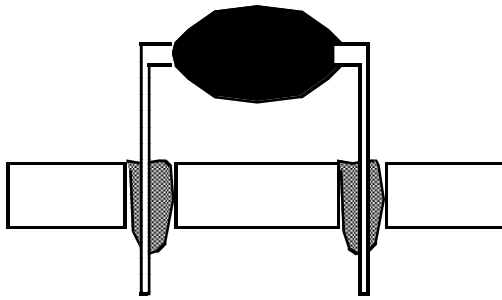
# Interconnect level 1 die-to-package

- Flip-Chip Bonding (倒装焊)

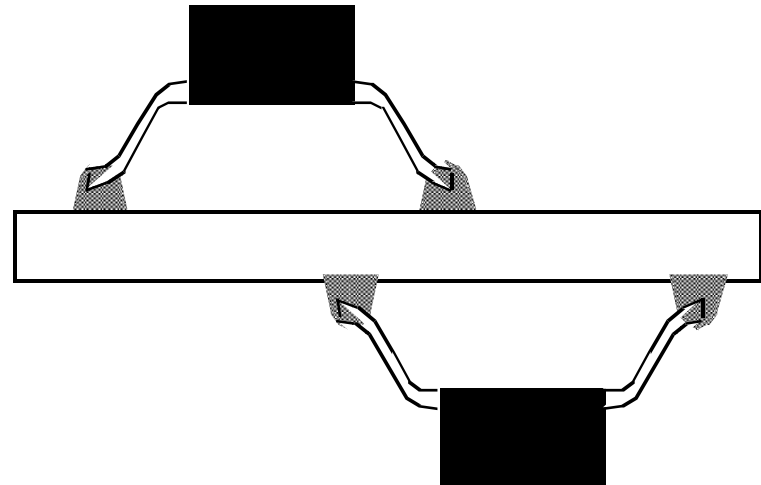




## Interconnect level 2 Package-Substrate-to-Board



(a) Through-Hole Mounting



(b) Surface Mount

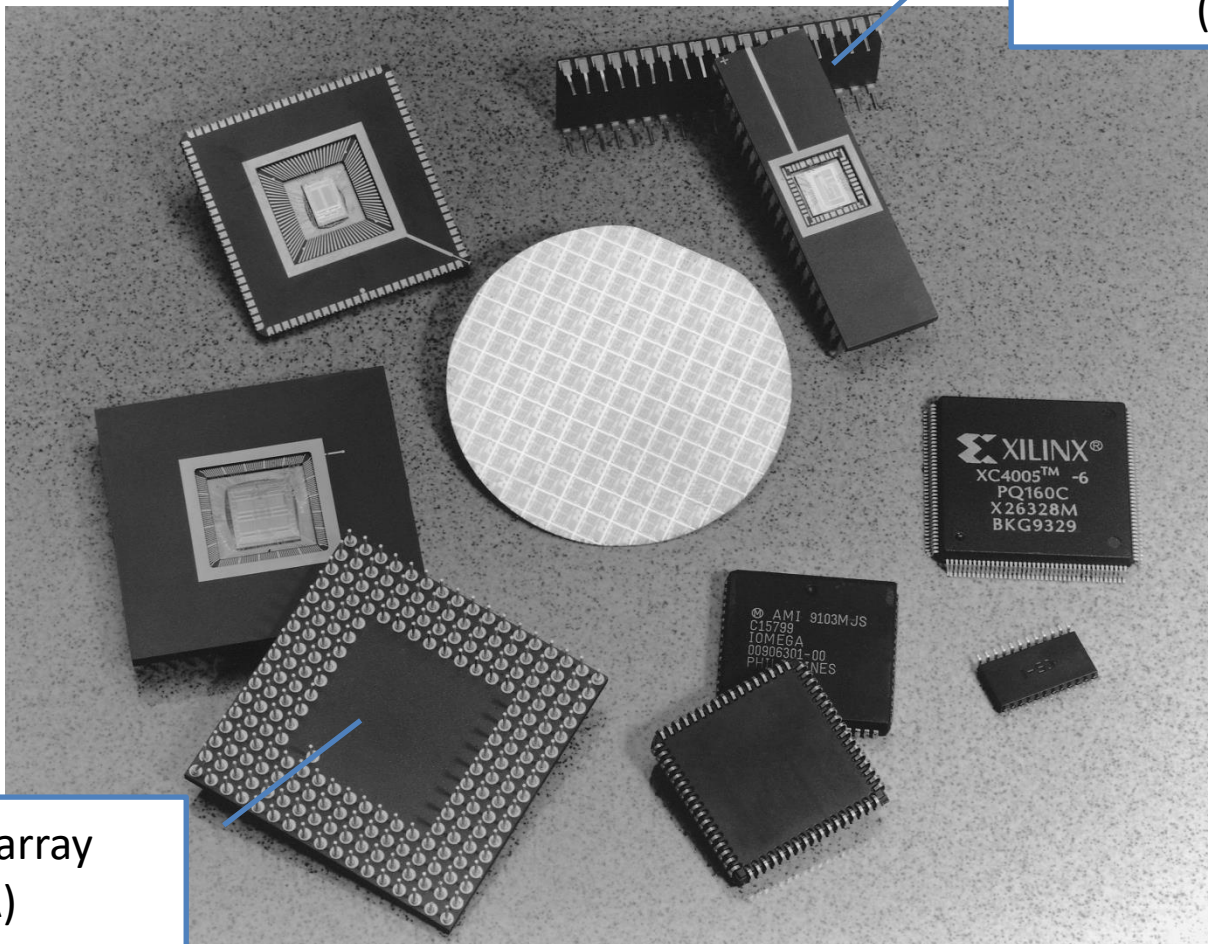




# Interconnect level 2

## Package-Substrate-to-Board

- Package Types



Dual-in-line package  
(DIP)

Pin-grid-array  
(PGA)



# LGA (Land Grid Array) 1366



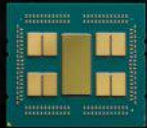




# More Advanced packaging

## PACKAGE ARCHITECTURES FOR CHIPLETS

MCM



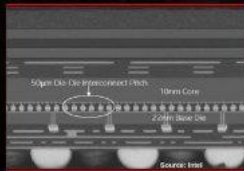
INFO-R



INFO-L

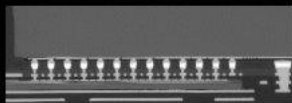


Foveros

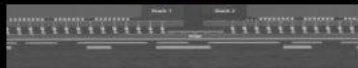


Intel: Foveros

EMIB



Co-EMIB



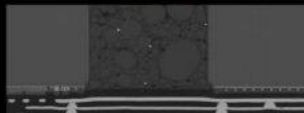
Intel: EMIB and Co-EMIB

CoWoS-L



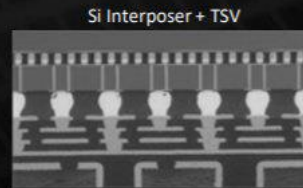
TSMC: INFO-R/-L, CoWoS-L

FoCoS



ASE: FoCoS

Micro Bump 3D



AMD Fiji GPU

INFO-POP



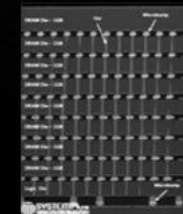
Apple A10 on FO+POP

Courtesy: SystemPlus Consulting



Courtesy: TechSearch

W2W stacking+ TSV+uBump



Samsung: HBM2 memory  
Courtesy: SystemPlus Consulting

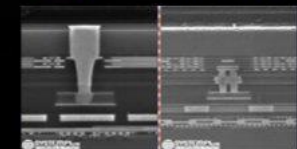
Foveros-ODI



Intel: Omni-directional Interconnect

WoW

W2W F2F di-electric bonding+ TSV      W2W F2F hybrid bonding w/o TSV



Sony: CMOS Image Sensors  
Courtesy: SystemPlus Consulting

No single package architecture works for all products - choice based on product PPAC



# Intel's EMIB (Embedded Multi-die Interconnect Bridge)

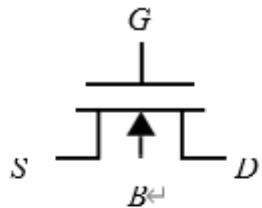


# Devices

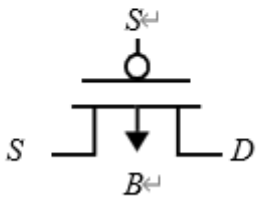


# The MOS Transistor

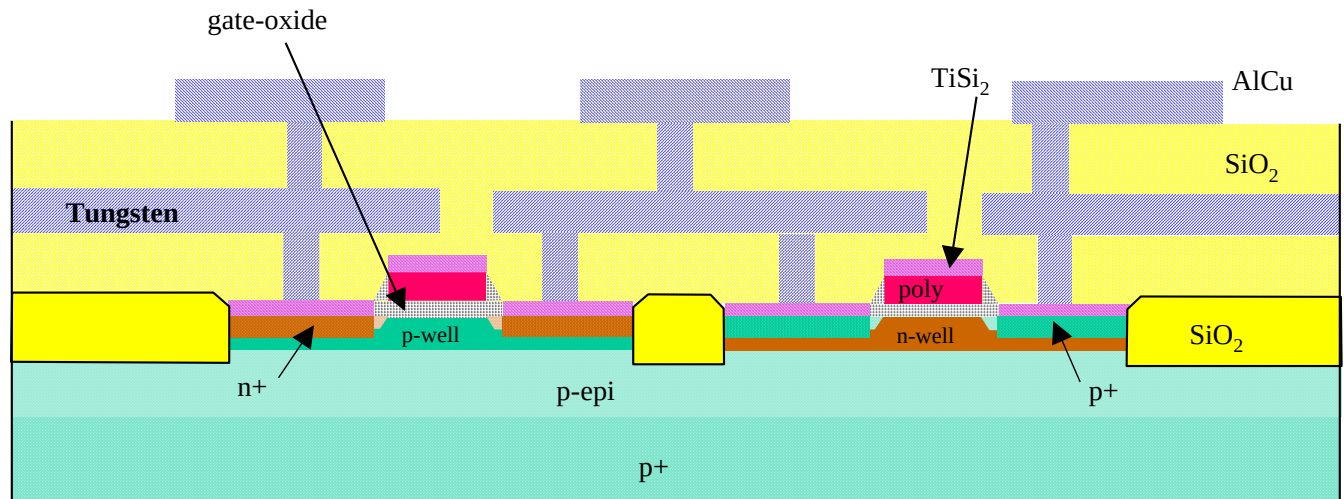
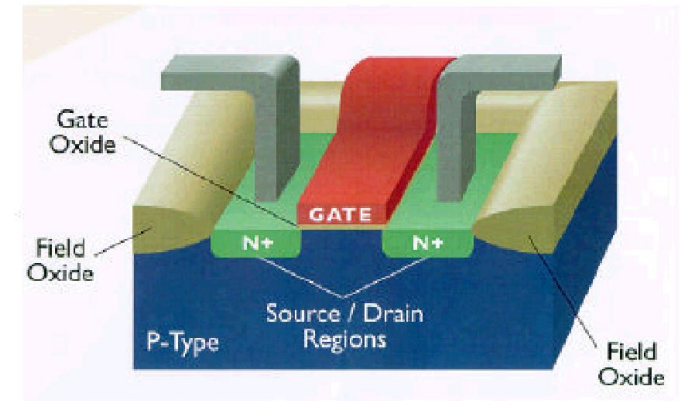
- Metal-oxide-semiconductor field-effect (金属氧化物半导体场效应管)



(a) NMOS transistor as 4-terminal device

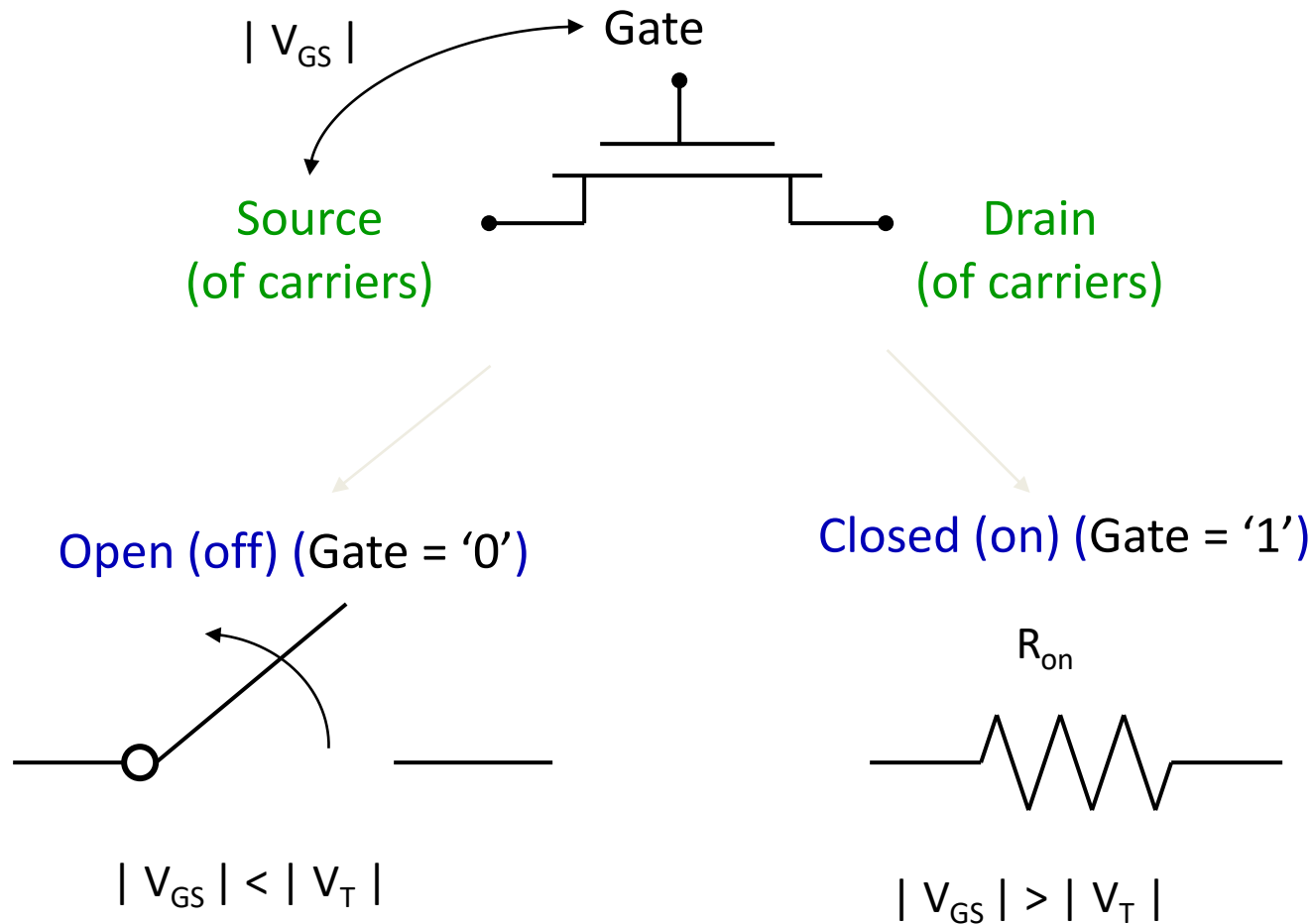


(a) PMOS transistor as 4-terminal device



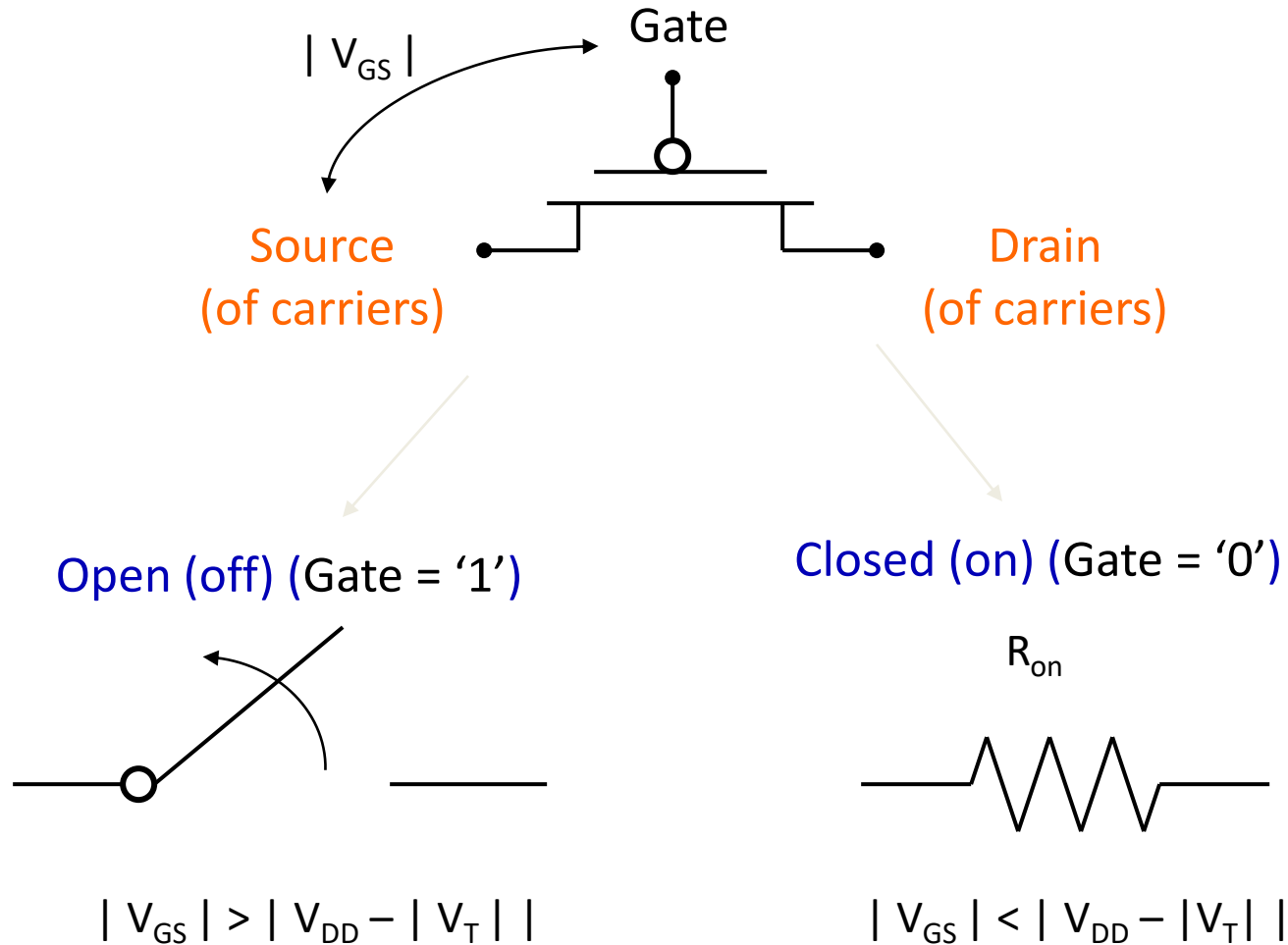


# Switch Model of NMOS Transistor





# Switch Model of PMOS Transistor

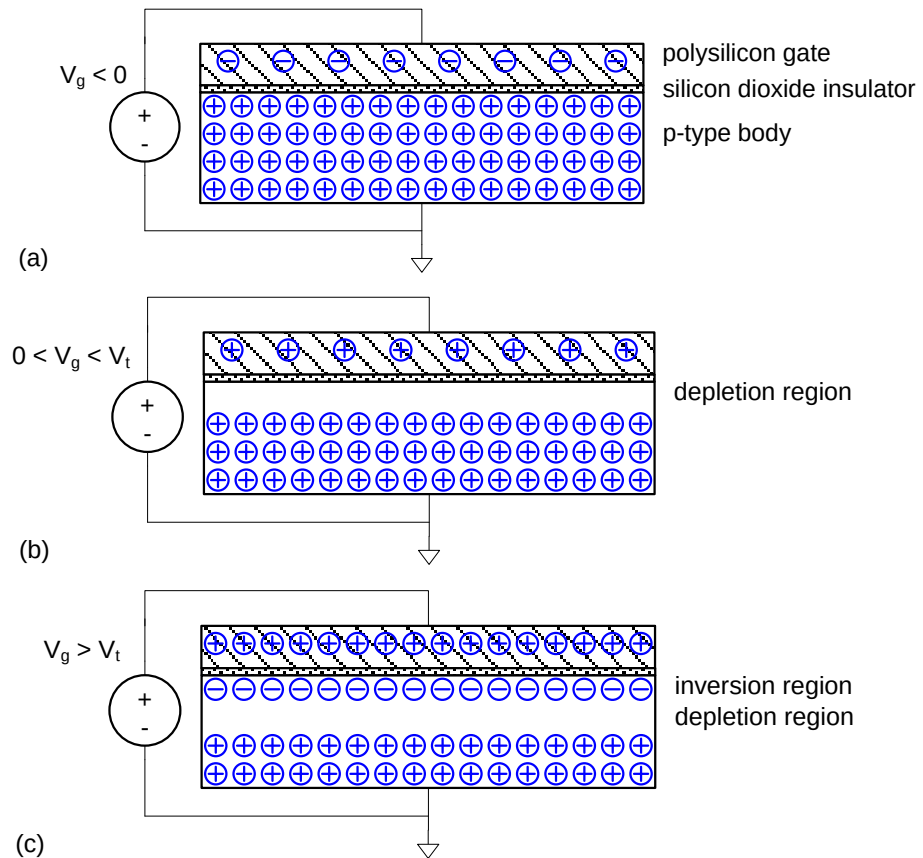






# MOS Capacitor

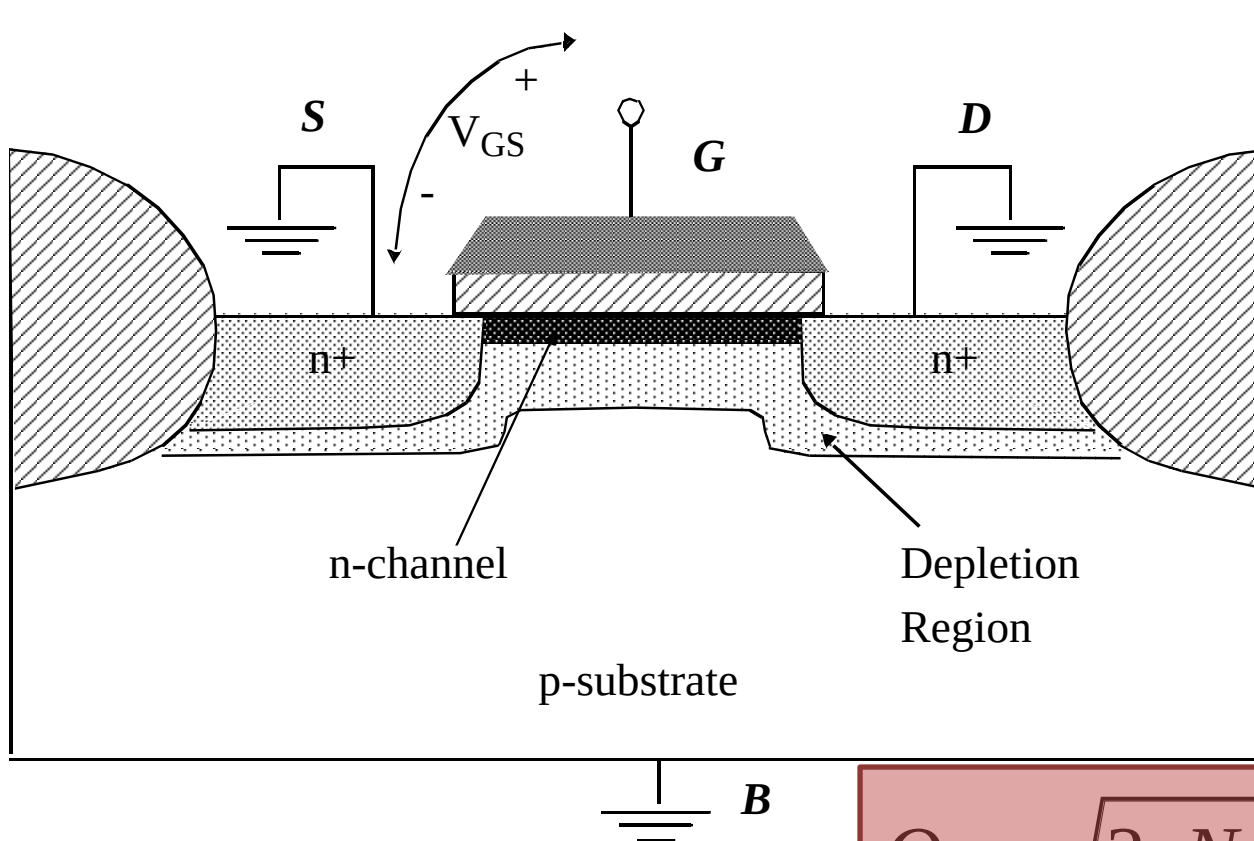
- Gate and body form MOS capacitor
- Operating modes
  - Accumulation
  - Depletion
  - Inversion





# Threshold Voltage

- 4-terminal



$$W_d = \sqrt{\frac{2\epsilon_{si}\phi}{qN_A}}$$

$$Q_d = \sqrt{2qN_A\epsilon_{si}\phi}$$

$$\phi_F = \phi_T \ln\left(\frac{N_A}{n_i}\right)$$

$$Q_B = \sqrt{2qN_A\epsilon_{si}(|(-2)\phi_F + V_{SB}|)}$$



# The Threshold Voltage

$$V_T = V_{T0} + \gamma(\sqrt{|(-2)\phi_F + V_{SB}|} - \sqrt{|2\phi_F|})$$

$V_{T0}$  is the **threshold voltage at  $V_{SB} = 0$**  and is mostly a function of the manufacturing process

Difference in work-function between gate and substrate material, oxide thickness, Fermi voltage, charge of impurities trapped at the surface, dosage of implanted ions, etc.

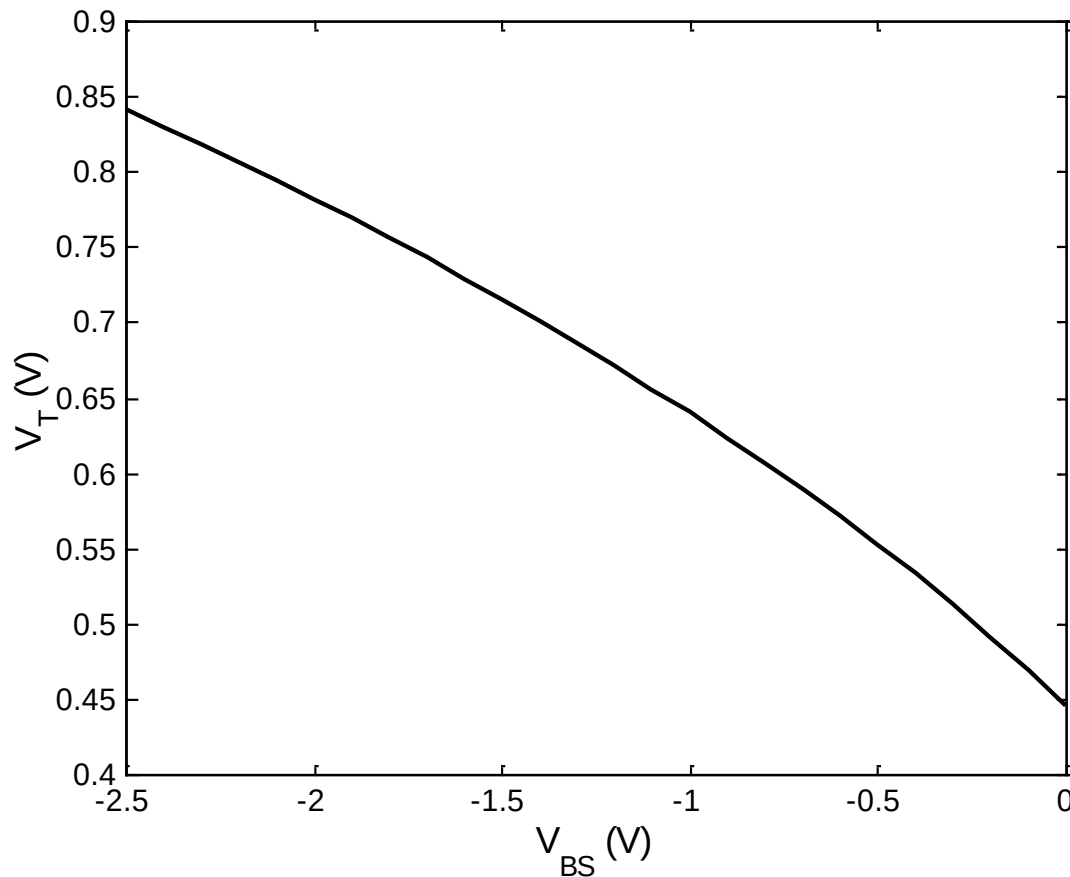
$V_{SB}$  is the **source-bulk voltage**

$\phi_F = -\phi_T \ln(N_A/n_i)$  is the **Fermi potential** ( $\phi_T = kT/q = 26\text{mV}$  at 300K is the thermal voltage;  $N_A$  is the acceptor ion concentration;  $n_i \approx 1.5 \times 10^{10} \text{ cm}^{-3}$  at 300K is the intrinsic carrier concentration in pure silicon)

$\gamma = \sqrt{(2q\epsilon_{si}N_A)/C_{ox}}$  is the **body-effect coefficient** (impact of changes in  $V_{SB}$ ) ( $\epsilon_{si} = 1.053 \times 10^{-10} \text{ F/m}$  is the permittivity of silicon;  $C_{ox} = \epsilon_{ox}/t_{ox}$  is the gate oxide capacitance with  $\epsilon_{ox} = 3.5 \times 10^{-11} \text{ F/m}$ )



# The Body Effect

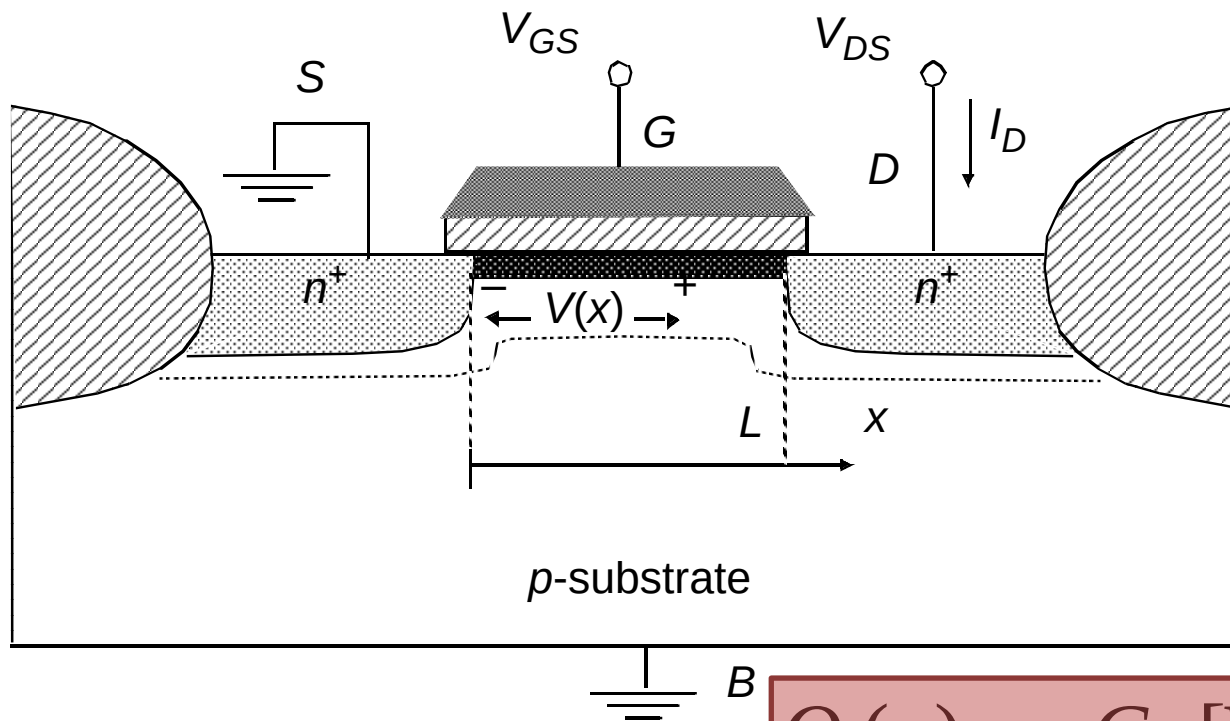


$V_{SB}$  is the substrate bias voltage (normally positive for n-channel devices with the body tied to ground)

A bias causes  $V_T$  to increase from 0.45V to 0.85V



# Resistive Operation



$$V_{GS} > V_T$$

$$V_{GS} - V_T > V_{DS}$$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

$$Q_i(x) = -C_{ox} [V_{GS} - V(x) - V_T]$$

MOS transistor and its bias conditions



# Resistive Operation

$$I_D dx = \mu_n C_{ox} W (V_{GS} - V - V_T) dV$$

$$I_D = k'_n \frac{W}{L} \left[ (V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right] = k_n \left[ (V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

- Process transconductance parameter

$$k'_n = \mu_n C_{ox} = \mu_n \frac{\epsilon_{ox}}{t_{ox}}$$

$\mu_n$  is the carrier mobility ( $\text{m}^2/\text{Vsec}$ )

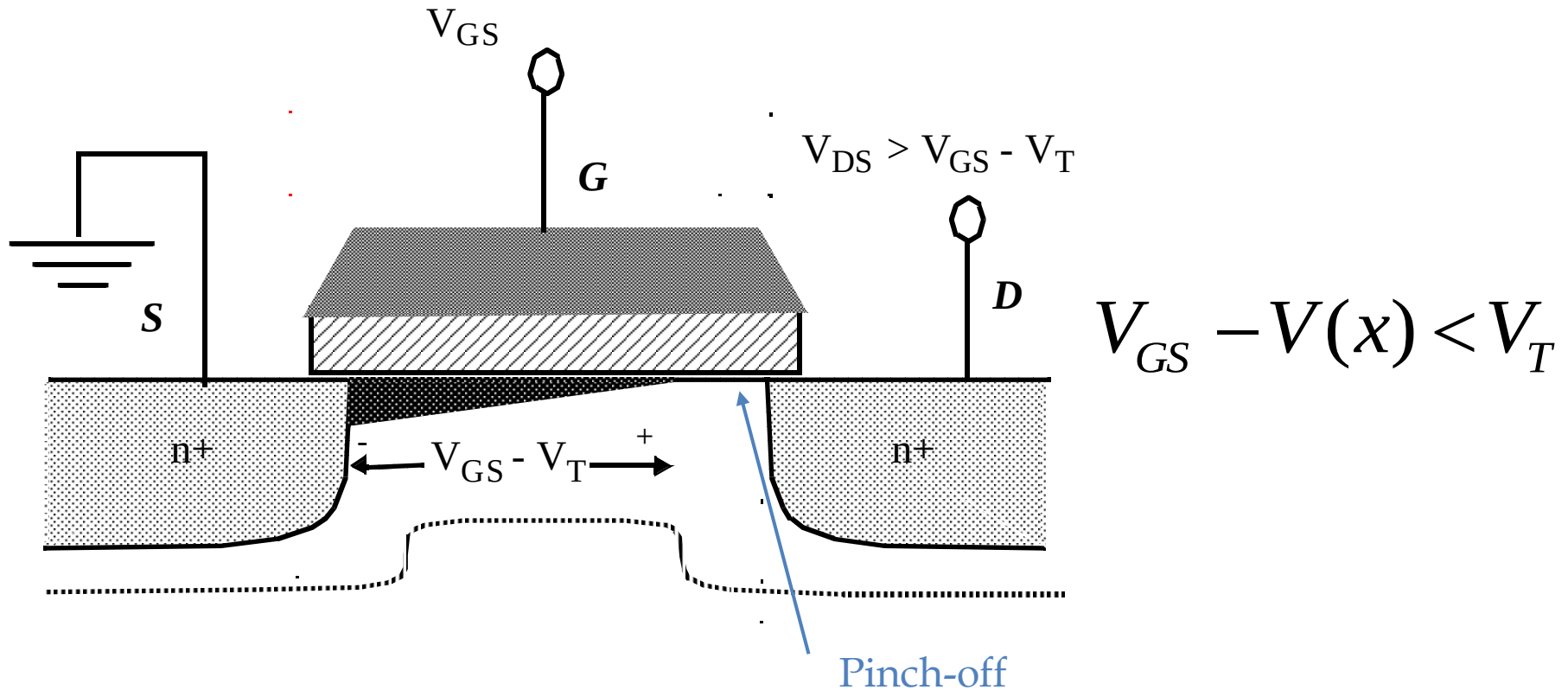
$k_n = k'_n W/L$  is the **gain factor** of the device

For long-channel devices ( $L > 0.25$  micron)

For small  $V_{DS}$ , there is a linear dependence between  $V_{DS}$  and  $I_D$ , hence the name **resistive** or **linear** region



# Saturation Region





# Saturation Region

the voltage difference over the induced channel (from the **pinch-off** point to the source) remains fixed at  $V_{GS} - V_T$

$$I_D = \frac{k'_n}{2} \frac{W}{L} (V_{GS} - V_T)^2$$

However, the effective length of the conductive channel is modulated by the applied  $V_{DS}$ , so

$$I_D = I_D' (1 + \lambda V_{DS})$$

where  $\lambda$  is the **channel-length modulation** (varies with the inverse of the channel length)





# Current-Voltage Relations Long-Channel Device

**Linear Region:  $V_{DS} \leq V_{GS} - V_T$**

$$I_D = k'_n \frac{W}{L} \left( (V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right)$$

with

$$k'_n = \mu_n C_{ox} = \frac{\mu_n \epsilon_{ox}}{t_{ox}} \quad \text{Process Transconductance Parameter}$$

**Saturation Mode:  $V_{DS} \geq V_{GS} - V_T$**

$$I_D = \frac{k'_n W}{2 L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

Channel Length Modulation  
↙

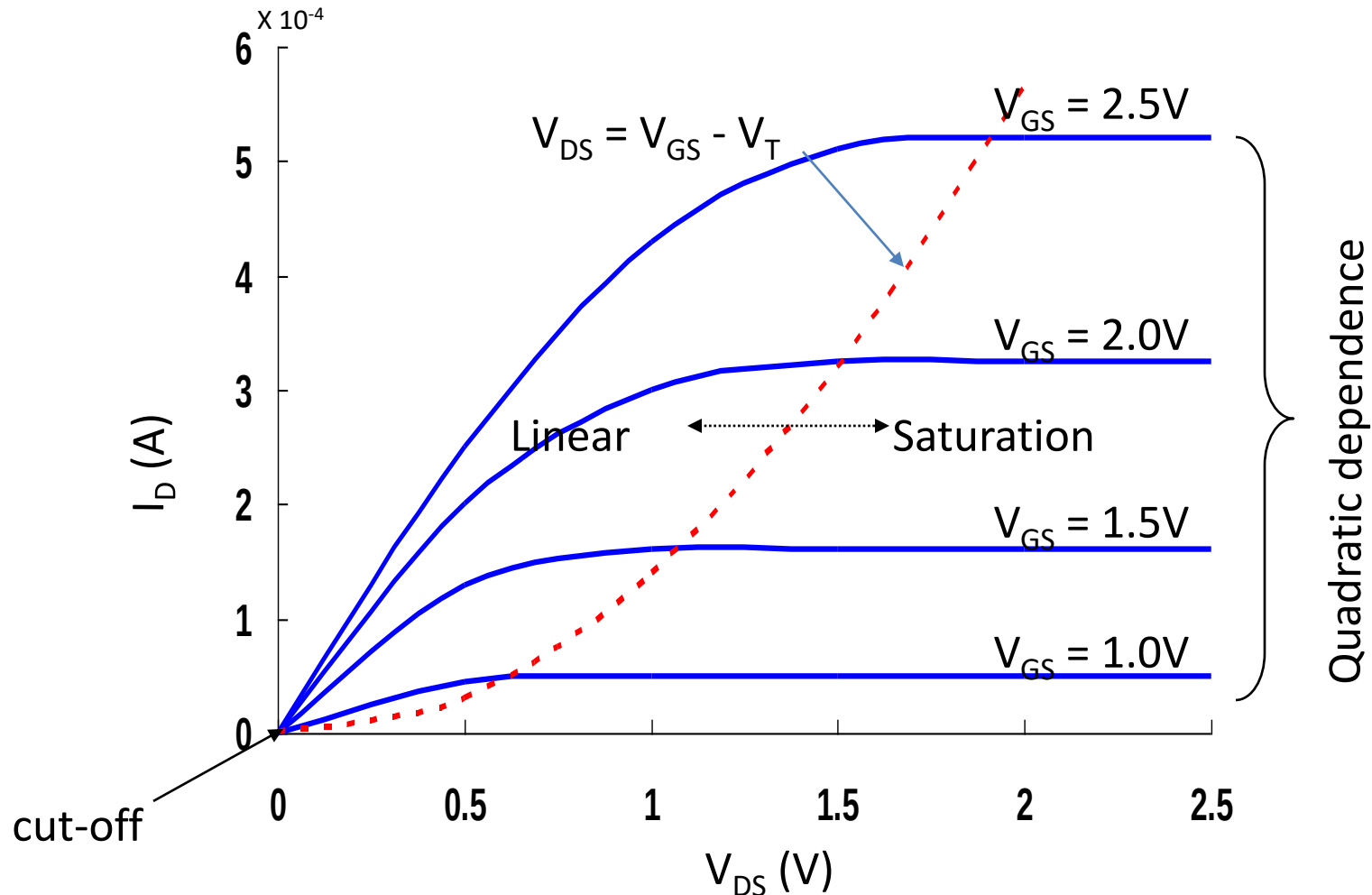


# Current Determinates

- For a fixed  $V_{DS}$  and  $V_{GS} (> V_T)$ ,  $I_{DS}$  is a function of
  - the distance between the source and drain –  $L$
  - the channel width –  $W$
  - the threshold voltage –  $V_T$
  - the thickness of the  $\text{SiO}_2$  –  $t_{ox}$
  - the dielectric of the gate insulator ( $\text{SiO}_2$ ) –  $\epsilon_{ox}$
  - the carrier mobility
    - for nfets:  $\mu_n = 500 \text{ cm}^2/\text{V-sec}$
    - for pfets:  $\mu_p = 180 \text{ cm}^2/\text{V-sec}$



# Long Channel I-V Plot (NMOS)

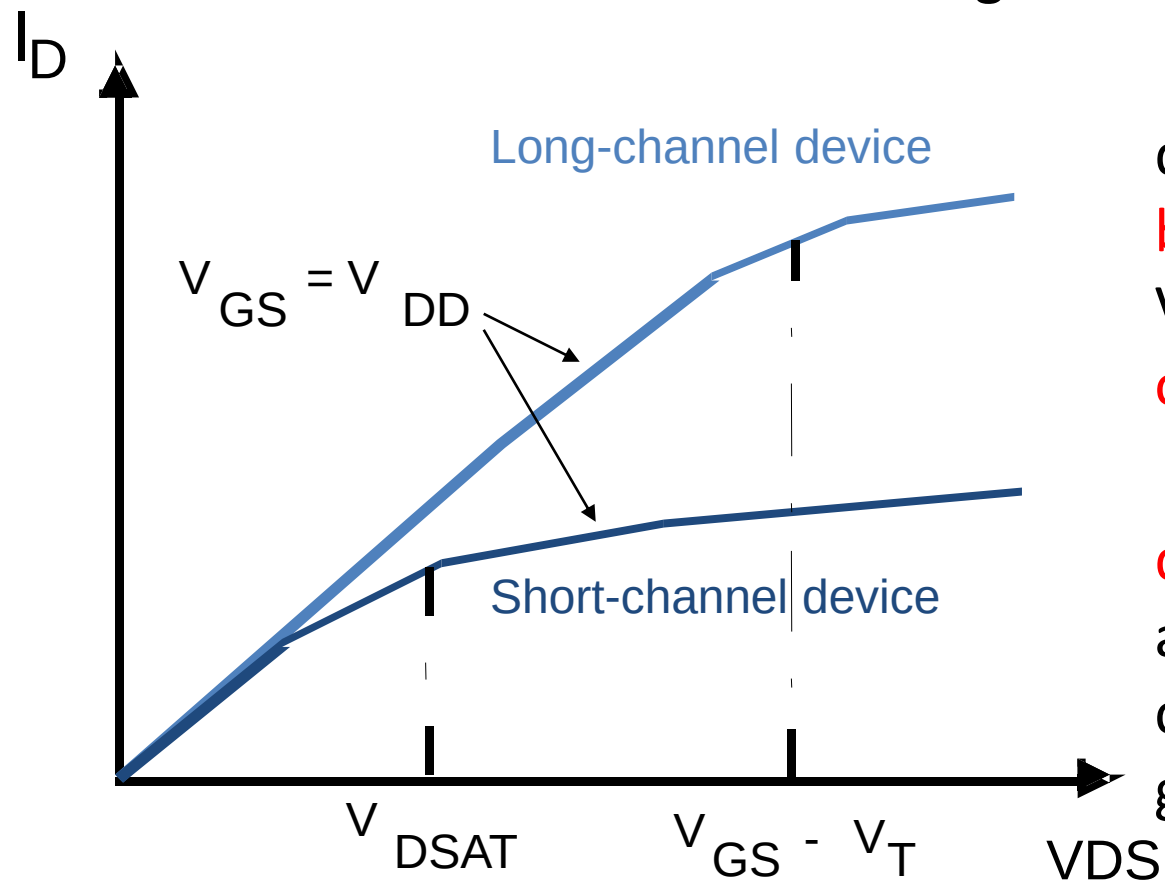


NMOS transistor,  $L_d = 10\mu m$ ,  $W/L = 1.5$ ,  $V_{DD} = 2.5V$ ,  $V_T = 0.4V$



# Velocity Saturation Effects

For short channel devices and large enough  $V_{GS} - V_T$

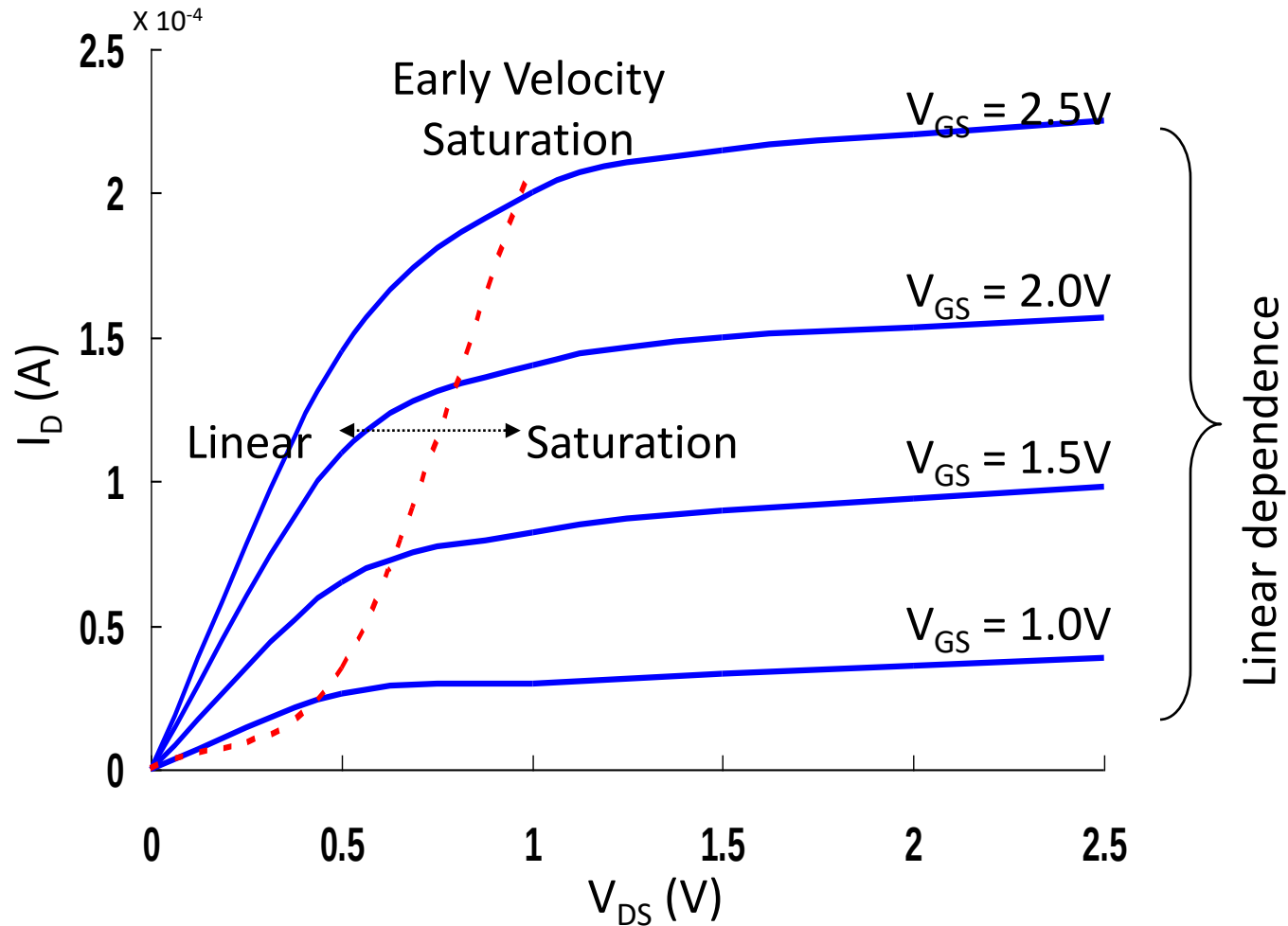


$V_{DSAT} < V_{GS} - V_T$  so the device enters saturation **before**  $V_{DS}$  reaches  $V_{GS} - V_T$  and operates **more often in saturation**

$I_{DSAT}$  has **a linear dependence** wrt  $V_{GS}$  so a reduced amount of current is delivered for a given control voltage



# Short Channel I-V Plot (NMOS)

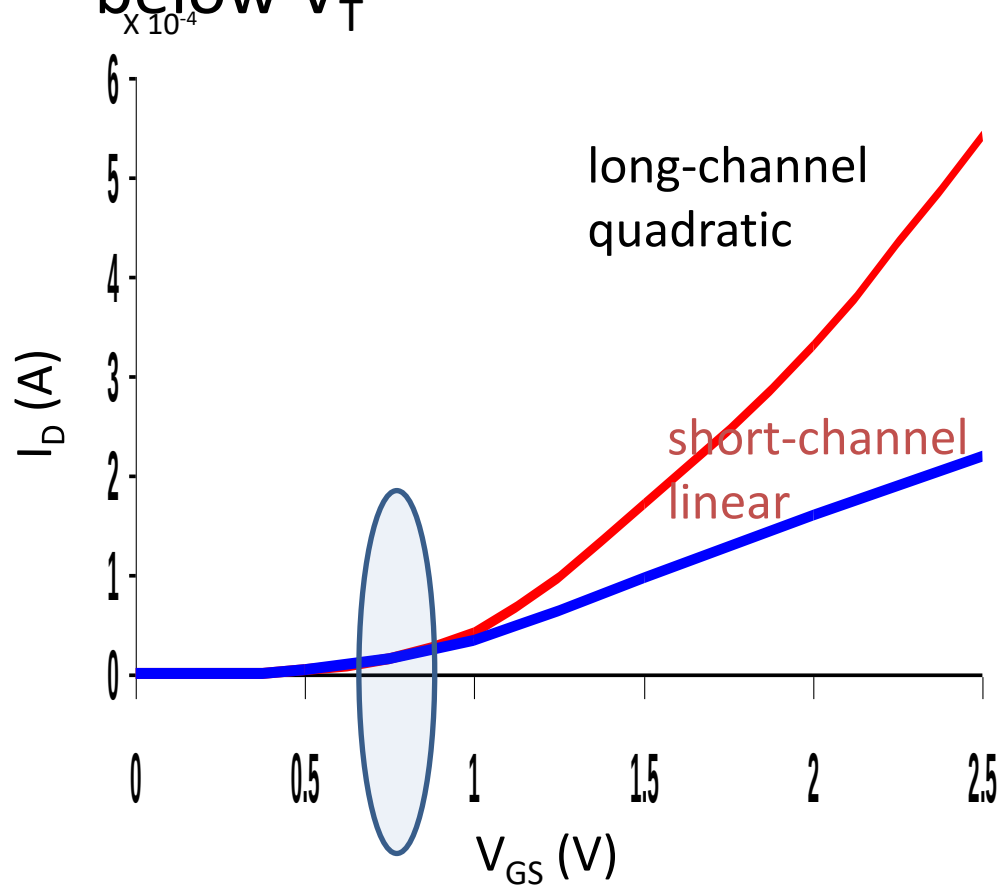


NMOS transistor,  $L_d = 0.25\mu m$ ,  $W/L = 1.5$ ,  $V_{DD} = 2.5V$ ,  $V_T = 0.4V$



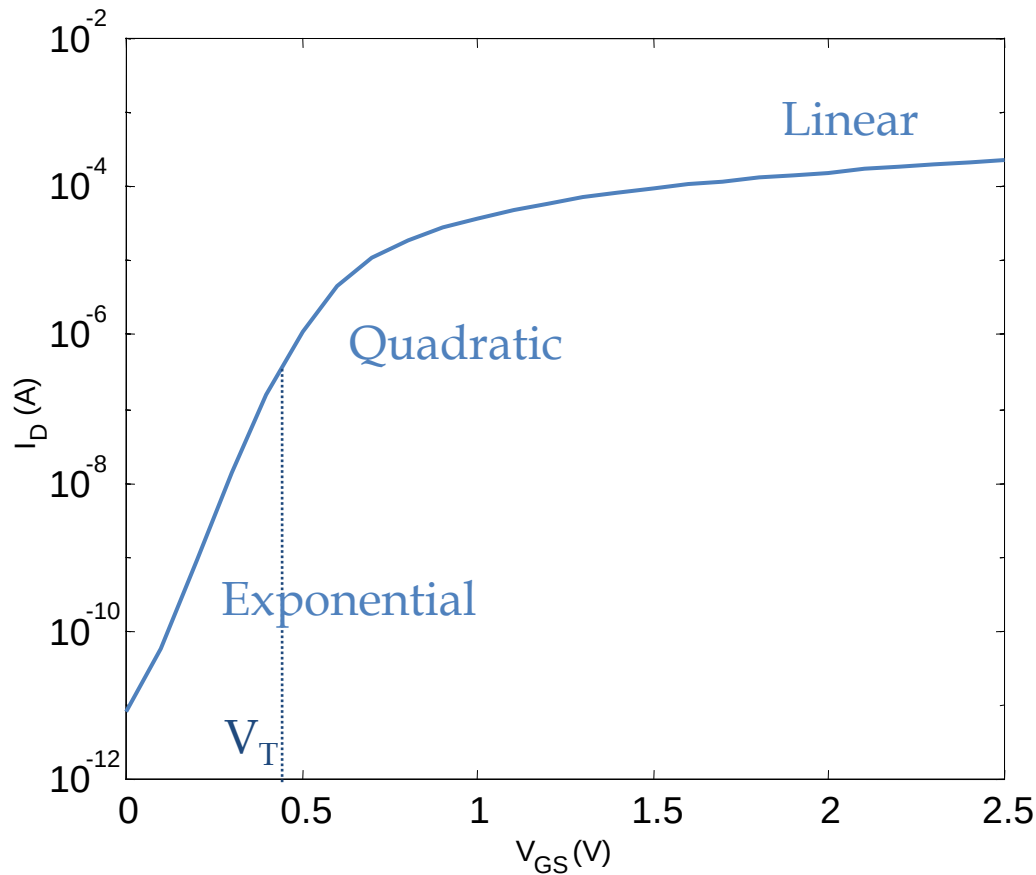
# Subthreshold conduction

- Transistor is already partially conducting for voltages below  $V_T$





# Subthreshold conduction



## The Slope Factor

$$I_D \sim I_0 e^{\frac{qV_{GS}}{nkT}}, \quad n = 1 + \frac{C_D}{C_{ox}}$$

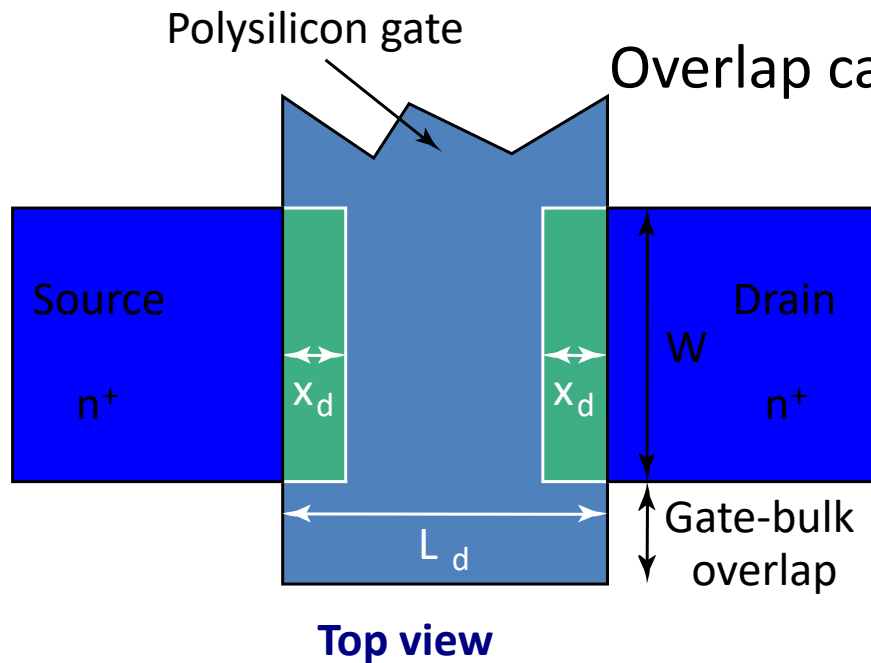
$S$  is  $\Delta V_{GS}$  for  $I_{D2}/I_{D1} = 10$

$$S = n \left( \frac{kT}{q} \right) \ln(10)$$

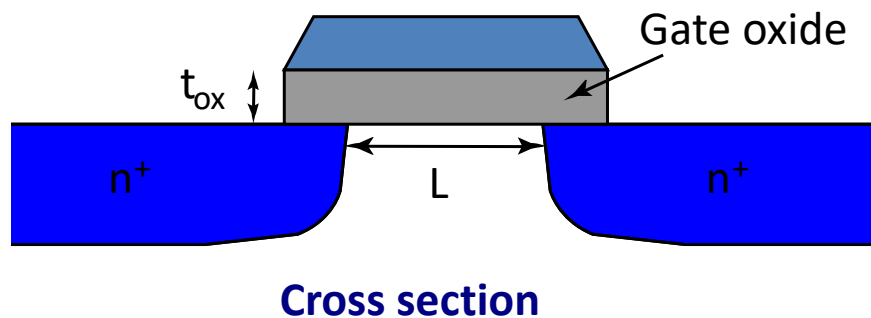
Typical values for  $S$ :  
60 .. 100 mV/decade



# MOS Structure Capacitances



$$C_{GSO} = C_{GDO} = C_{ox} x_d W = C_o W$$

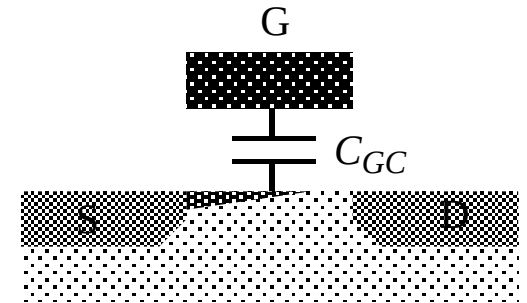
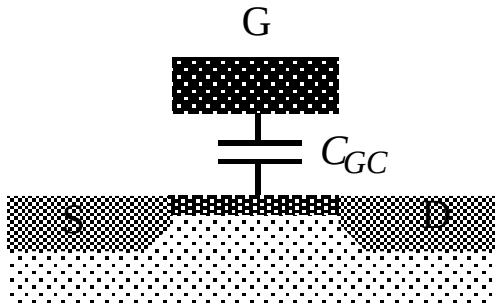
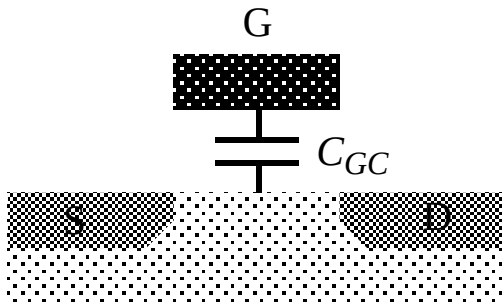






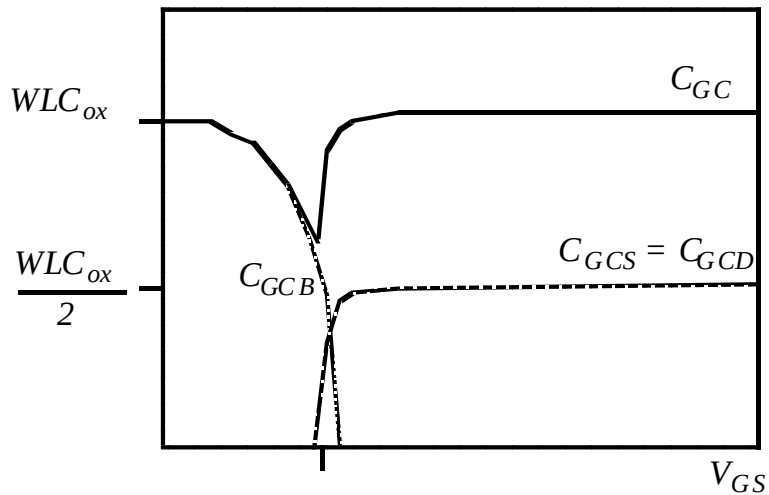
# MOS Channel Capacitances

- The gate-to-channel capacitance depends upon the operating region and the terminal voltages

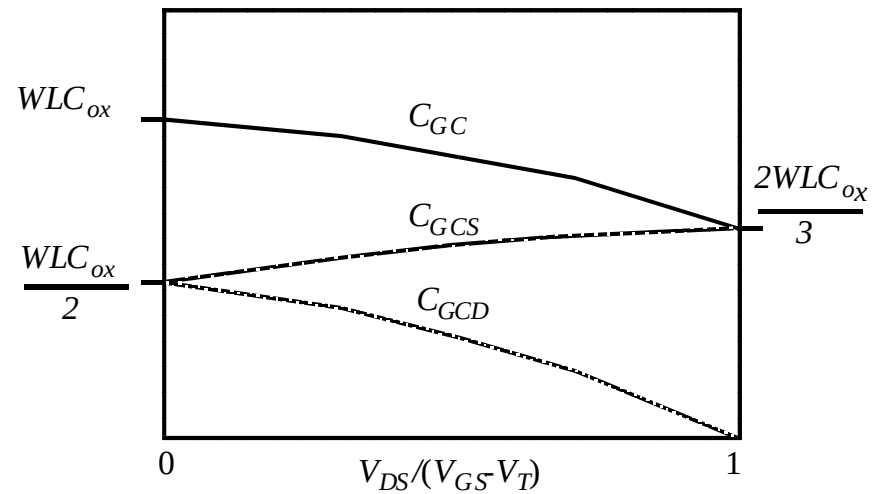




# MOS Channel Capacitances



Capacitance as a function of  $V_{GS}$   
(with  $V_{DS} = 0$ )



Capacitance as a function of the  
degree of saturation



# Average Distribution of Channel Capacitance

Channel capacitance components are nonlinear and vary with operating voltage

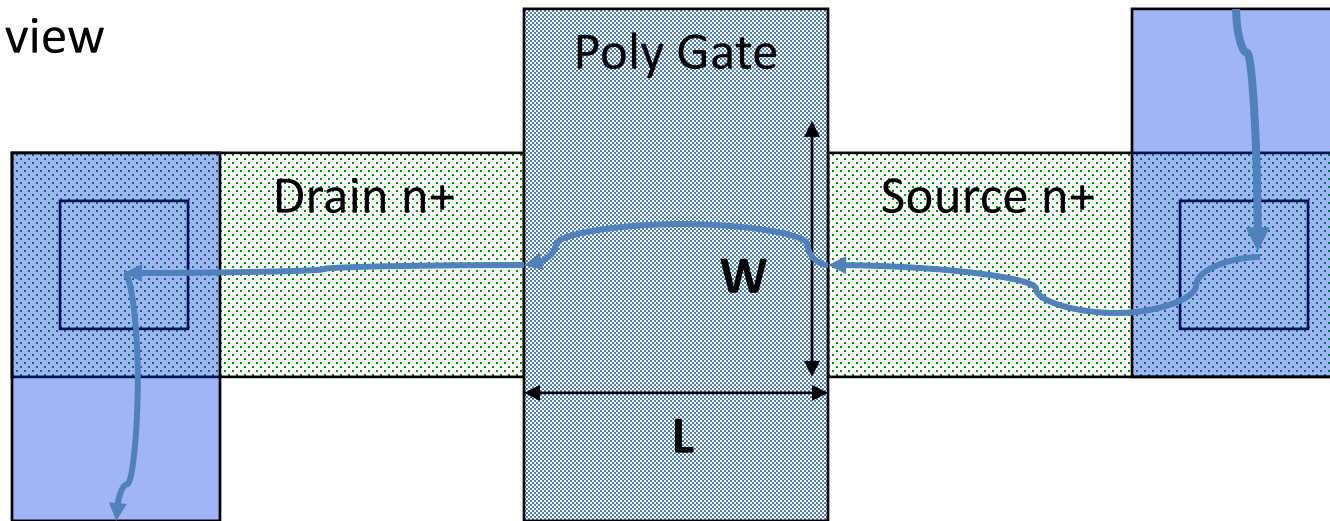
Most important regions are cutoff and saturation since that is where the device spends most of its time

| Operation Region | $C_{GCB}$  | $C_{GCS}$       | $C_{GCD}$    | $C_{GC}$        | $C_G$                   |
|------------------|------------|-----------------|--------------|-----------------|-------------------------|
| Cutoff           | $C_{ox}WL$ | 0               | 0            | $C_{ox}WL$      | $C_{ox}WL + 2C_0W$      |
| Resistive        | 0          | $C_{ox}WL/2$    | $C_{ox}WL/2$ | $C_{ox}WL$      | $C_{ox}WL + 2C_0W$      |
| Saturation       | 0          | $(2/3)C_{ox}WL$ | 0            | $(2/3)C_{ox}WL$ | $(2/3)C_{ox}WL + 2C_0W$ |



# Sources of Resistance

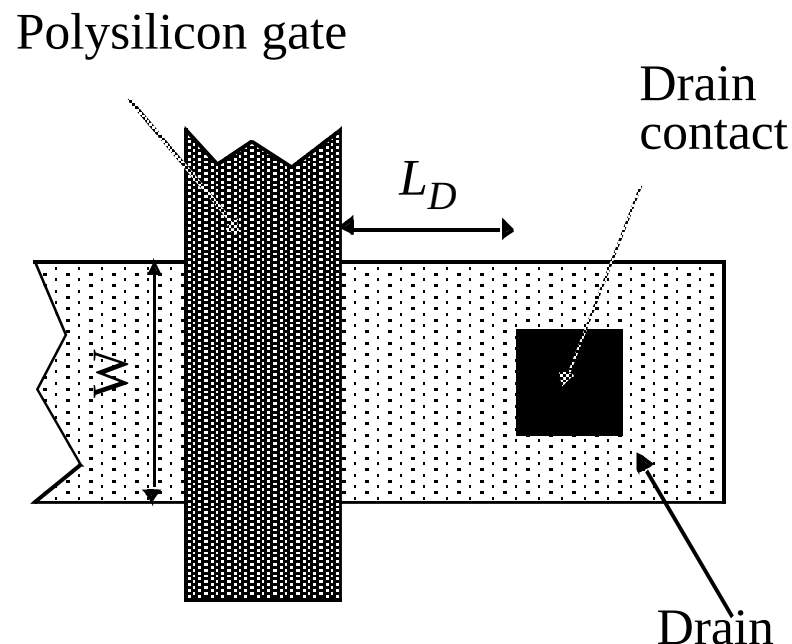
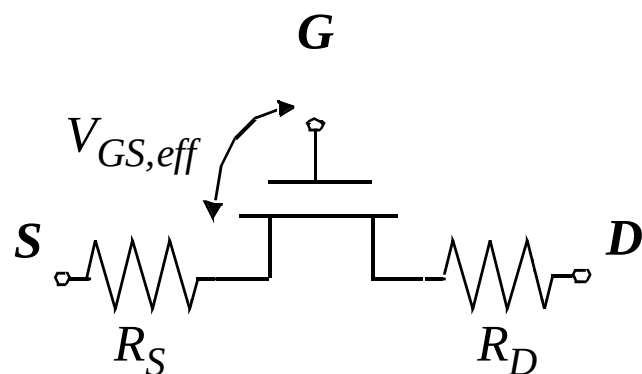
Top view



- MOS structure resistance -  $R_{on}$
- Source and drain resistance
- Contact (via) resistance
- Wiring resistance



# Source and Drain Resistance



$$R_{S,D} = (L_{S,D}/W)R + R_C$$

where  $L_{S,D}$  is the length of the source or drain diffusion  
 $R$  is the sheet resistance(薄层电阻) of the source or drain diffusion (20 to 100  $\Omega/\square$ )



# Summary

- Overview of manufacturing process
  - Introduce manufacturing steps in IC based on lithography tech.
  - Introduce design rules and IC packaging tech.
- Intuitive understanding of device operation
  - basic device equations for operating regions
  - MOS Structure Capacitances